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## Abstract

Owing to the striking properties, perovskite materials have clutched the interest in the field of optoelectronics. The property of being band tunable via composition alteration makes the perovskites fit in many optoelectronic applications. However, the compositional instability of mixed halide perovskites (MHPs) under illumination or by charge-carrier injection, intercepting the exploitation of mixed halide strategy which effectively tailors the perovskite's bandgap. Herein, the segregation of halide ions in lattice promotes the formation of individual halide-rich domains that severely influence the electronic band structures of the MHPs. In this scenario, extensive research is heading to perceive the mechanisms that drive the halide segregation and abate the process accordingly. Since the impact of ion segregation directly determines the performance of the devices, in-depth understanding is a prerequisite. This perspective provides a compressive understanding of different aspects such as structural origin, influence of light intensity, temperature, and atmospheric conditions that root the segregation process. A rigorous analytical discussion on approaches to play with cations and anions compositions, dimensionality, and Lewis base treatment that are detrimental for the halide

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segregation process has been presented. This perspective contains a critical view on the halide segregation process that paves the route for future research.

## 1. Introduction

Halide perovskite's crystal structure has a general chemical formula  $ABX_3$  where A is a monovalent cation, B is a divalent metal cation and X is a monovalent anion mostly halide.<sup>[1]</sup>  
<sup>2]</sup> The B-site elements octahedrally coordinated with halides to form  $BX_6$  inorganic frameworks and the A components occupy the cuboctahedral cavity formed by nearest-neighbor X atoms in  $AX_{12}$  polyhedrons (Figure 1a). Fully inorganic perovskites are known since the end of the 19th century,<sup>[3]</sup> and their hybrid organic-inorganic counterparts  $MAPbX_3$  and  $FAPbX_3$  (methylammonium (MA)-  $CH_3NH_3^+$ ; formamidinium (FA)-  $CH(NH_2)_2^+$ ) are known since the late 1970s.<sup>[4, 5]</sup> However, from the last decade, perovskite materials have engrossed researchers owing to their inherent optoelectronic and photovoltaic properties.<sup>[6-11]</sup> In an optoelectronic context, perovskites have received considerable attention in the 1990s as channel layers for field-effect transistors and as active layers in light-emitting diodes (LEDs).<sup>[12-14]</sup> Only a decade later, perovskites have drawn major research significance as an absorber layer in highly efficient photovoltaic (PV) devices. With certified efficiency of >25% as of today, perovskites have proven themselves as a preferable replacement for Si-based devices.<sup>[15, 16]</sup> Having highly diversified dimensions such as quantum dots, nanowires, 2D layers, quasi 2D, and bulk 3D they became prominent in photovoltaic technology. If we carefully analyze the efficiency enhancement of solar cell devices, though the  $MAPbI_3/FAPbI_3$  single-phase materials exhibiting the highest efficiency,<sup>[15-18]</sup> they are intrinsically unstable to moisture, oxygen, and UV light that attributes to the spontaneous exothermic reaction as a result of energy change for the phase decomposition. The kinetic barrier of these materials may prevent the decomposition to a certain extent, but to make this technology commercialize, it is indispensable to enhance the stability without compromising the efficiency.<sup>[19]</sup> On the other side, among all the striking

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properties of the perovskites, bandgap tunability is one of the fascinating properties of perovskites that needs to be explored more in-depth to use in the tandem structures. The promising method to tackle both the stability and bandgap tunability can be achieved via partial replacement of the native ions in the lattice of the best performing MAPbI<sub>3</sub> with the similar ions such as I<sup>-</sup> with Br<sup>-</sup> or Cl<sup>-</sup>, Pb<sup>2+</sup> with Sn<sup>2+</sup>, and CH<sub>3</sub>NH<sub>3</sub><sup>+</sup> with FA<sup>+</sup> or inorganic Cs<sup>+</sup> or Rb<sup>+</sup>. Presently, the most performant perovskite-based optoelectronic devices are conceived by employing mixed components in view of their enhanced stability with respect to their single-phase counterparts.<sup>[20, 21]</sup> Some MHPs with optimum compositions have exhibited improved efficiencies in solar cells are shown in figure 1b.<sup>[22-29]</sup> This shows that future trends for perovskite photovoltaic technology are moving towards perovskites with mixed components. Interest in these systems originates from a conjunction of attractive semiconductor characters, includes: strong light absorption capacities, electronic structures offering direct tunable bandgaps, readily accessible synthesis of high-quality films that enable facile structure-property-correlation, and rapid device prototyping/optimization.<sup>[30-33]</sup>

To date, many strategies were demonstrated to vary the bandgap by incorporating different pertinent elements in the lattice sites of the perovskite. Moreover, it is a facile procedure to exchange elements in perovskite lattice that significantly influence their optoelectronic properties. In the case of A-site cations, though they do not contribute directly towards the band structure. They play a significant role in providing structural stability via charge compensation owing to their electrostatic (van der Waals) interactions with the BX<sub>6</sub> octahedra cage.<sup>[2]</sup> Their ionization energies and electron affinities do not exhibit a uniform trend. Nevertheless, any change in the size of A component can either contract or expand the crystal lattice which consequently varies the optical properties of the perovskite.<sup>[23, 34-36]</sup> The hybridization between the B and X states in crystal structure reduces slightly because of deformations such as octahedral tilting and distortion of the octahedra in the BX<sub>3</sub> framework.

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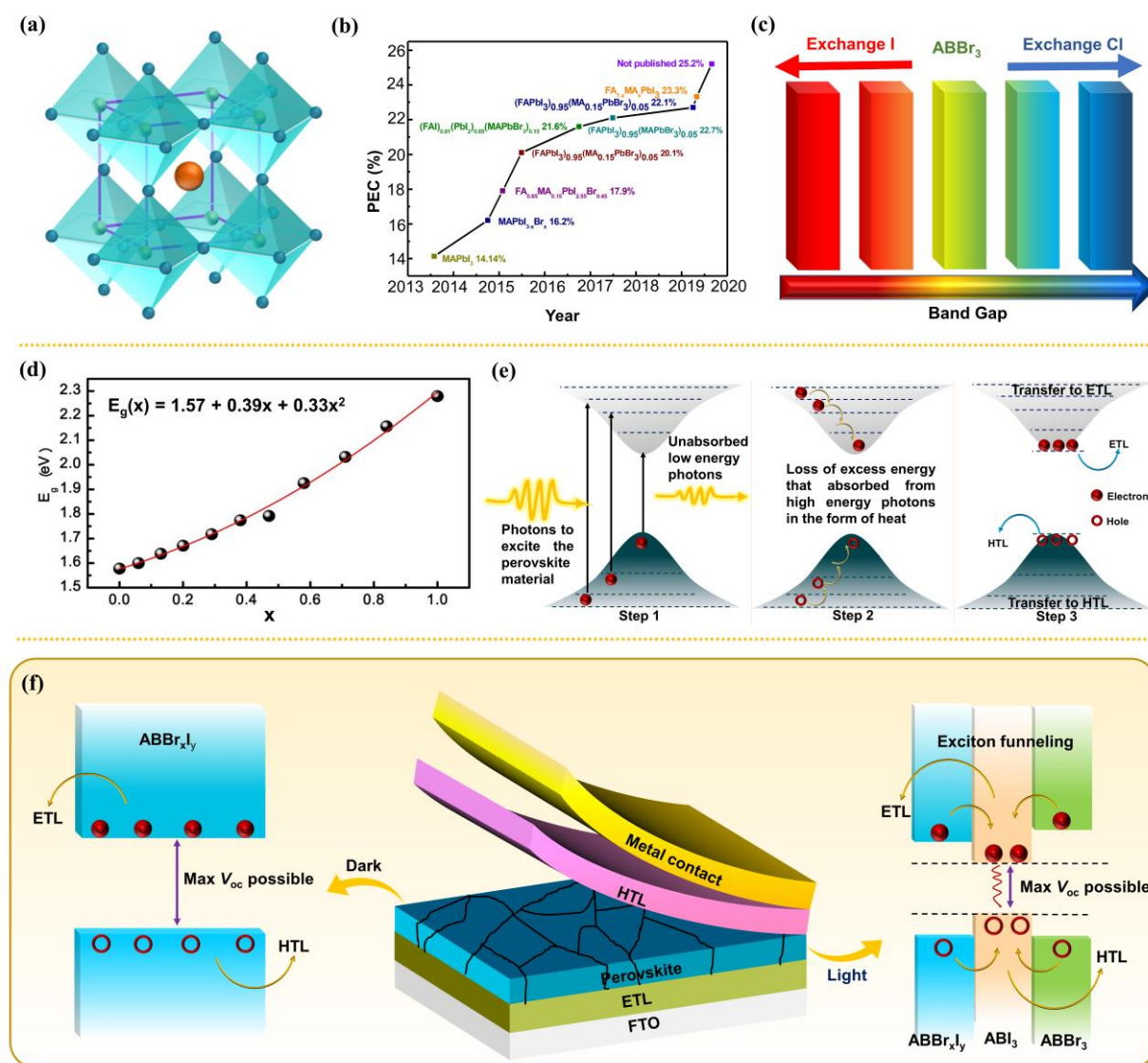
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It brings down both the conduction band minimum (CBM) and valence band maximum (VBM), however, VBM (ionization energy) is influenced more because of its high dependence on hybridization. On the other hand introduction of larger cations expands the lattice and lowers the levels of B. Here also, both CBM and VBM go down (increase in electron affinity and ionization energy), but now CBM (electron affinity) is impacted more due to its high dependence on B levels. Therefore, structure deformation and volume expansion cause the rise in ionization energy and electron affinity but distortion effect ionization energy most whereas volume expansion effect electron affinity.<sup>[37]</sup>

Similarly, alloying of the B cations also alter the optoelectronic properties of perovskites.<sup>[38]</sup> The B cation sites in hybrid perovskites are occupied by the group IVA metals in a divalent oxidation state. In general, upon proceeding upwards in group IVA of the periodic table (Pb → Ge) atomic levels shift upwards that is consistent with the electronegativity. Herein, replacing  $\text{Pb}^{2+}$  with  $\text{Sn}^{2+}$  both the ionization energy and electron affinity of the material decreases, whereas splitting *s* and *p* states in small atoms is less than that of Pb atom's splitting, which results in a larger upward shift of the *s* level than *p* level that leads to a more upward shift in VBM than that of CBM. The upward shift of VBM is further pronounced by a slight enhancement in B,*s*/*X*,*p* hybridization. Therefore, shifts of CBM and VBM promote the reduction of the Sn-based perovskites bandgaps.<sup>[37]</sup> Howbeit with lower atomic number group IVA elements, stability stands as a major curb as the stability of +2 oxidation state for each element goes on deteriorating due to reduced inert electron pair effect.<sup>[8]</sup> The material with formula  $\text{MASnI}_3$  possesses a more optimal bandgap (1.2—1.4 eV) than  $\text{MAPbI}_3$  (1.5—1.8 eV) but suffers from rapid degradation due to the inherent instability of the  $\text{Sn}^{2+}$  state as it readily oxidizes to  $\text{Sn}^{4+}$  upon interaction with air producing a volatile  $\text{SnI}_4$  compound.<sup>[39]</sup>

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**Figure 1.** Schematic representation of (a) Perovskite structure. (b) Efficiency chart of the perovskites-based solar cell.<sup>[22-29]</sup> (c) Evolution of the bandgap tuning in MHPs with anion variation. (d) Quadratic relationship of the MAPb(I<sub>1-x</sub>Br<sub>x</sub>)<sub>3</sub> band-gaps as a function of Br composition (x). Reproduced with permission.<sup>[30]</sup> Copyright 2013 American Chemical Society. (e) Influence of perovskite material bandgap on harvesting the photons of different energies. (f) Effect of phase segregation on the performance of photovoltaic devices.

Similar to the incorporation of different cations in the A and B sites of the perovskite lattice structure, halides such as **chloride** (Cl<sup>-</sup>), **bromide** (Br<sup>-</sup>), and **iodide** (I<sup>-</sup>) have been incorporated into pure MAPbI<sub>3</sub>, FAPbI<sub>3</sub>, inorganic halide perovskites, or mixed-cation

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(FA/MA, FA/MA/Cs, FA/Cs, FA/MA/Cs/Rb, etc.) lead halide perovskites.<sup>[21, 38, 40, 41]</sup> Herein,

mixed halides in the X-sites of perovskite lattices impart a dramatic effect on electron affinity, ionization energy, stability, and carrier dynamics which sways the optoelectronic properties.<sup>[41-</sup>

<sup>47]</sup> Generally, upon proceeding down the group VIIA from Cl to Br to I, absorption onset of the spectra shift to longer wavelengths (Figure. 1c, 1d), for example, the absorption onset

wavelengths of MAPbCl<sub>3</sub>, MAPbBr<sub>3</sub>, and MAPbI<sub>3</sub> are in between 400-500, 500-600, and 700-

800, respectively. This attributes to a downward shift of the Pb,*p* atomic level (increase in

atomic size), which influence the CBM, and a considerably strong upward shift of the X,*p* level

(decrease in electronegativity of the halides) that is opposed by two weak factors 1) downshift

of the Pb,*s* level as discussed above for Pb,*p* atomic level 2) small reduction of the Pb,*s*/X,*p*

hybridization strength going from Cl to Br to I, which influence the VBM. This synergetic

movement of the conduction and valency bands leads to reduction in the bandgap from Cl to

Br to I.<sup>[37, 48]</sup> Among all halide perovskites, iodide-based hybrid perovskites are proven as

highly efficient materials because of their bandgaps in the range of 1.5—1.8 eV that is almost

perfect to use as a single junction solar cell absorber. As a matter of fact, iodide-based

perovskites have provided the foundation for hybrid perovskite counterparts, but their ease of

degradation has introduced quite a lot of stability concerns. It was reported that the partial

replacement of the I<sup>-</sup> with Br<sup>-</sup> to  $x = 0.2$  in MAPb(I<sub>1-x</sub>Br<sub>x</sub>)<sub>3</sub> has significantly improved the

material stability without deteriorating its performance. It was observed that the lattice constant

decreases in proportion to  $x$ , while the bandgap and the exciton binding energy increase. The

size effect on lattice parameters will induce compressive stress on Pb-I bonds causing structural

distortion that leads to the increase in the bandgap device, it is accompanied by downshifting

of VBM or increase in ionization energy as a result of stronger hybridization of the Br 4*p* states

with the Pb *s* states compared to I 5*p* states.<sup>[49, 50]</sup> Therefore, the materials can be tailored to the

ideal bandgap to use as a higher bandgap absorber in the dual absorber tandem. Noh et al. have

reported that increasing the Br content  $x$  in MAPb(I<sub>1-x</sub>Br<sub>x</sub>)<sub>3</sub> leads to phase transition from

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tetragonal to cubic phase at  $x = 0.13$ , while bandgap modification following the quadratic function  $E_g(x) = 1.57 + 0.39x + 0.33x^2$  (eV) of Br content. They observed considerable improvement in stability without affecting the performance for low Br concentrations ( $x < 0.2$ ).<sup>[30]</sup> In augmentation, alloying MHPs ( $\text{Br}^- / \text{I}^-$ ) with chloride ( $\text{Cl}^-$ ) has manifested improved stability, diffusion lengths, and carrier lifetimes.<sup>[51]</sup> The addition of chloride to iodide-based perovskites was also shown to push the efficiencies. However, it has been argued that as a result of the larger difference in ionic radii,  $\text{Cl}^-$  incorporation yields low miscibility with  $\text{I}^-$ .<sup>[44, 52, 53]</sup>

## 2. Carrier dynamics in MHPs affecting the performance of solar cells and LEDs

MHPs have become significant to harvest the solar spectrum efficiently in solar cells and to obtain the color pure tunable emitters for LEDs.<sup>[54]</sup> In the case of solar cells, short circuit current density ( $J_{sc}$ ) and open-circuit voltage ( $V_{oc}$ ) are the crucial parameters that decide device performance, Where the  $J_{sc}$  is the current through the solar cell when the voltage across the solar cell is zero. At most moderate resistive losses in an ideal solar cell, the short-circuit current and the light-generated current are identical. Therefore, the  $J_{sc}$  is the largest current that may be drawn from the solar cell. It is determined by the number of photons absorbed and charge collection. On the other hand,  $V_{oc}$  is the voltage at zero current, which is the maximum voltage available from a solar cell. It is determined by the difference between the fermi level of electron transport material and the highest occupied molecular orbital of hole transport materials, as the consequence of their fermi level alignment in the heterojunctions of the solar cell architecture. If the active perovskite material is of a small bandgap, then it can absorb a significant range of photons in the solar spectrum to obtain a high  $J_{sc}$ . However, a considerable amount of excess energy from high-energy photons gets dissipated in the form of heat that causes a drop in  $V_{oc}$ . On the other hand, where perovskites having a higher bandgap can harvest only those photons

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with higher energy that bestows high  $V_{oc}$  by letting low energy photons unabsorbed and leads to the limitation of  $J_{sc}$  (Figure 1e). Therefore, there is a need to trade-off between  $J_{sc}$  and  $V_{oc}$  to obtain high power conversion efficiency. It is worth mentioning that,  $V_{oc}$  is limited by the bandgap of active material, as the matter of fact maximum attainable  $V_{oc}$  value cannot be more than the bandgap of the active perovskite material. In general, the CBM of the wide range of the perovskites fairly aligns with the  $\text{TiO}_2$  conduction band as there is not much difference of electron affinity among them but the ionization energy of the HTL should be low enough to match with the lowered ionization energies of the  $\text{Br}^-$  incorporated perovskites to avoid loss in  $V_{oc}$ . In addition, HTM must have low electron affinity to avoid the recombination of electrons and holes.<sup>[55]</sup> Similarly, for LEDs, there is a need for high phase pure materials with a wide range of bandgaps to obtain specific color emissions with significant color purity. Hence, there is an indispensable need for material that possesses the incredible property to give high efficiency along with easy tunability to attain a wide range of band gaps.

As mentioned above, MHPs are synthesized mainly to exploit the bandgap tunability character of the perovskites, thereby  $V_{oc}$  and  $J_{sc}$  can be optimized in single junction and tandem solar cells, and pure color emission tunability can be achieved in LEDs. However, the major disadvantage of MHPs remains in their long-term stability that predominantly stems from light-induced and bias-induced halide segregation.<sup>[56-60]</sup> Under continuous illumination or charge carrier injection, MHPs undergo reversible phase segregation. Here,  $\text{MAPbI}_3$  with bandgap 1.6 eV and  $\text{MAPbBr}_3$  with bandgap 2.3 eV have the highest achievable  $V_{oc}$  of 1.1 eV and 1.5 eV, respectively. However, the intermediated combinations are not in parallel to the predicted outputs where  $\text{MAPb}(\text{I}_x\text{Br}_{1-x})_3$  ( $1 > x > 0.2$ ) got stuck at the achievable  $V_{oc}$  of 1.1 eV, which is equivalent to  $V_{oc}$  of  $\text{MAPbI}_3$ , that attributes to the phase segregations process.<sup>[30]</sup> The segregated state of the film has iodide-rich domains with low bandgap and bromide-rich domains with high bandgap. In general, charge carriers tend to funneling spontaneously to the lower bandgap material region from the higher one. Therefore, all the charge carriers generated



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funnel to the low bandgap iodide-rich domains, where bromide-rich domains become photoluminescence (PL) silent and neither contribute to the enhancement of  $V_{oc}$  in solar cells nor leads to color pure high-frequency emitters for LEDs.<sup>[60, 61]</sup> Hoke et al. are the first to report  $\text{MAPb}(\text{Br}_{0.2}\text{I}_{0.8})_3$  perovskites phase-segregate into Br and I-rich domains.<sup>[60]</sup> Numerous investigations have reproduced the effect of establishing its ubiquity across different mixed-halide systems and even structural morphologies.<sup>[62-64]</sup> In MHPs having I and Br, the segregated I-rich domains act as recombination centers for free charge carriers because of their band position and hence block electron flow into a solar device (Figure 1f). This segregation process has a significant effect on the optical and electrical properties of MHPs.<sup>[64]</sup> Therefore, red shifting of the PL emission does limit the color design in LEDs too. Hence, the minimization of halide segregation in MHPs is a prerequisite to optimize the performance of the optoelectronic devices fabricated with these materials. Most perplexing is the reversible nature of the segregation process. Upon removal of illumination, the segregated phases prefer to re-mix and form the MHPs. Presently, it is widely accepted that ion migration is one of the causes for anomalous photocurrent hysteresis phenomena.<sup>[65-68]</sup> Even after considerable research on the halide segregation phenomenon, many facets of the process are still debated in the literature and a clear picture of the underlying fundamental mechanisms is yet to be drawn. Several chronicles of research on MHPs have provided hints on the segregation process such as the influence of compositions, polarons, defects, grain sizes, and crystallinity of MHPs. In augmentation, studies have ascertained the external factors such as temperature, illumination conditions, and the surrounding environment of MHPs affecting their stability. Furthermore, theoretical models have been developed so far as to discover the cause of phase segregation in MHPs.<sup>[68-70]</sup>

When the research is focused on preventing halide segregation, Mahesh et al. have demonstrated that losses in  $V_{oc}$  of MHPs are more due to non-radiative recombinations than losses associated with the formation of phase-segregation.<sup>[59]</sup> Motti et al. have addressed this

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by deploying a combination of and optical-pump terahertz-probe (OPTP) and photoluminescence spectroscopy to study the relation between charge-carrier dynamics and phase segregation in MHPs. They have reported that halide segregation results in a trivial change in THz effective charge carrier mobilities, and the charge carriers are not strongly localized within the I-rich domains. Further, in heterogeneous MHP films, they demonstrated enhanced phonon anharmonicity in the I-rich domains that supports ionic migration. This anharmonicity provides evidence for picosecond charge funneling into the narrow-bandgap I-rich domains that foster charge carrier radiative recombination in phase segregated MPHs. In photovoltaic materials efficient radiative recombinations are a good sign, however, high radiative rates in the phase segregated perovskite can lead to higher losses in MPHs.<sup>[71]</sup> Therefore, it is the need of the hour to understand and control the recombination dynamics in the MPHs rather than heading only to suppress the halide segregation.

### 3. Composition variation

Hoke et al. have demonstrated structural change under illumination using powder X-ray diffraction (XRD) (Figure 2a) and PL measurements (Figure 2b).<sup>[60]</sup> Splitting of XRD diffraction peaks with light soaking and re-formation of the original sharp diffraction patterns under darkness was observed. Since perovskites with higher bromide content have smaller lattice constants than those with higher iodide content, the splitting of XRD diffraction peaks by illumination is consistent with the presence of a minority phase with higher iodide content ( $x \sim 0.2$ ) and a majority phase with higher bromide content ( $x \sim 0.7$ ) compared to the original material. Meanwhile, the spectral change that occurred over time under illumination in the PL measurement showed that the initial emission peaks from the mixed-phase decreased while two new different signals emerged. It has been described that the newly obtained low-energy and high-energy peaks correspond to the segregated I-rich and Br-rich domains, respectively. Also, after the photoirradiation, the spectral features from segregated domains are bleached and

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returned to the original peak of the mixed phase. This recovery process takes longer than the initial segregation process by several minutes to an hour.<sup>[72]</sup> Misra et al. have investigated photo-stability of  $\text{MAPbI}_{3-x}\text{Br}_x$  ( $x = 0.11, 0.16, 0.22, 1$ ) under concentrated sunlight (100 suns).<sup>[73]</sup> Herein,  $\text{MAPbBr}_3$  was found to be stable exhibiting no degradation, whereas increasing iodine incorporation leads to accelerated photochemical degradation yielding  $\text{PbI}_2$  as a final remaining product that decomposes into metallic  $\text{Pb}(0)$  and  $\text{I}_2$ .<sup>[74]</sup> It is worth mentioning that no degradation was recorded for  $\text{MAPbI}_{3-x}\text{Br}_x$  under shelf-storage conditions, but degradation was observed to occur under illumination conditions. Later on, numerous studies have shown similar behavior for perovskites having different cations and anions proving the ubiquitous nature of the light/bias induced phase-segregation process.<sup>[57, 62]</sup>

### 3.1. Variation and alloying of cations:

Halide segregation has been observed in MHPs possessing different ions at A and B sites of the lattice. However, alloying of cations was shown to increase the photostability of perovskites.<sup>[60, 75-79]</sup>  $\text{FA/Cs}^+$  and  $\text{FA/MA/Cs}^+$  perovskites are observed to be significantly more stable against halide segregation compared to only MA-based perovskites. Also, the composition range over which phase segregation occurs is variable with the cations used. While  $\text{MAPb}(\text{Br}_x\text{I}_{1-x})_3$  has been found to segregate for halide ratios  $0.2 < x < 1$ ,<sup>[60, 80]</sup> its pure FA and  $\text{Cs}^+$  counterparts exhibit comparable halide segregation only within the narrower approximate compositional ranges  $0.55 < x < 0.91$ <sup>[79]</sup> and  $0.4 < x < 1$ ,<sup>[81]</sup> respectively thus presenting slightly improved stability ranges. Moreover, triple cation alloying at the A site of MHPs showed much improved stability. Although alloying of A-site cations showed an increase in phase stability significantly, the reason behind the improved photostability against phase segregation upon A cation alloying is not clear yet.

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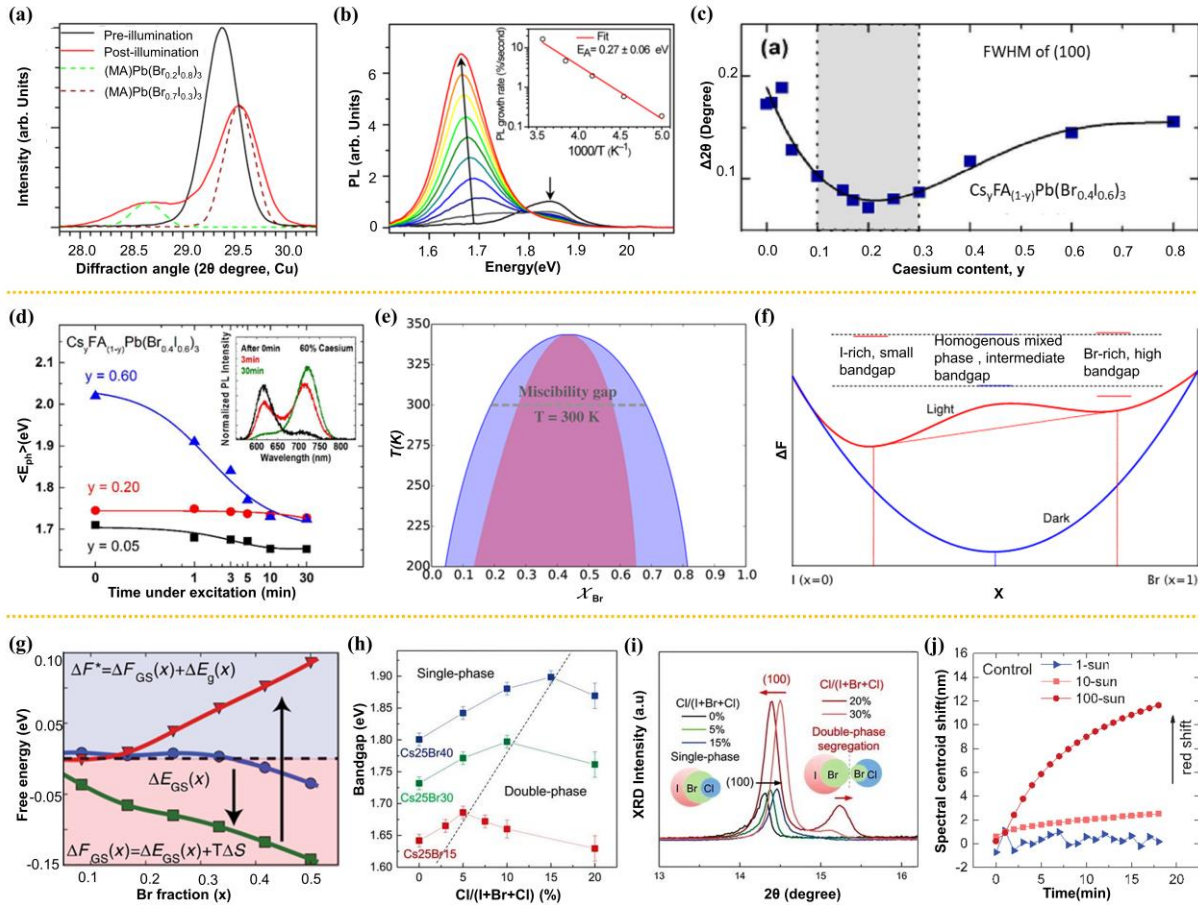
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Rehman et al. have investigated the influence of  $\text{Cs}^+$ /FA cations composition on the crystallinity of the  $\text{Cs}_y\text{FA}_{(1-y)}\text{Pb}(\text{Br}_{0.4}\text{I}_{0.6})_3$  perovskites. As shown in figure 2c, incorporation of  $\text{Cs}^+$  in the  $\text{FAPb}(\text{Br}_{0.4}\text{I}_{0.6})_3$  leads to the reduction of FWHM, and a minimum was reached at  $y=0.2$ . Since a decrease in XRD peak width is associated with an increase in crystallite size and crystallinity. Photostability check of  $\text{Cs}_y\text{FA}_{(1-y)}\text{Pb}(\text{Br}_{0.4}\text{I}_{0.6})_3$  perovskites for different values of  $y$  (Figure 2d) revealed that these perovskites are stable towards segregations at  $y=0.2$ . Reduced photostability was noticed for both lower and higher compositions. They proposed that appropriate  $\text{Cs}^+$  or FA content in the crystals makes redistribution of halide ions thermodynamically unfavorable and hence suppresses phase segregation.<sup>[78]</sup> On the other hand, Beal et al. have synthesized a series of  $(\text{FA}_y\text{Cs}_{1-y})\text{Pb}(\text{Br}_x\text{I}_{1-x})_3$  MHPs by varying the ratio of FA to  $\text{Cs}^+$  cations at fixed  $\text{Br}^-/\text{I}^-$  ratio while maintaining the film morphology relatively constant in the range of composition. They demonstrated that with an increase in  $\text{Cs}^+$  content, MHPs become more photo-stable. In addition, they observed cubic to tetragonal transition with increasing  $\text{Cs}^+$  content. However, the same cannot be applied in MA/ $\text{Cs}^+$  system as these MHPs are not stable against phase segregation even at very high  $\text{Cs}^+$  composition despite having tetragonal geometry.<sup>[82]</sup> Hence, the phase of the perovskites film cannot be considered as the sole factor for improved stability of MHPs. Mathew et al. have shown a five-fold decrease in the segregation rate constant upon replacement of MA by  $\text{Cs}^+$  as A-site cation in MHPs.<sup>[83]</sup> The significant influence of  $\text{Cs}^+$  on slowing down the segregation rate can be attributed to the stronger binding of halide ions.<sup>[84-87]</sup> In addition, Alexander et al. have demonstrated the stability of  $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{Br}_{0.4}\text{I}_{0.6})_3$  over  $\text{MAPb}(\text{Br}_{0.5}\text{I}_{0.5})_3$  based on the ionic pathways.<sup>[88]</sup> Therefore, suppression of halide ion mobility through A-site cation substitution can indirectly enhances the photostability of perovskite films.<sup>[77, 89]</sup>

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Alloying B site cations also contributes towards the enhancement of MHP's photostability.<sup>[90, 91]</sup> Yang et al. have shown that substituting a fraction of the  $\text{Pb}^{2+}$  cations with smaller  $\text{Sn}^{2+}$  cations in  $\text{MASn}_{1-y}\text{Pb}_y\text{Br}_{3-x}\text{I}_x$  films enhanced its photo-stability.<sup>[90]</sup> An increase in the micro-strain of perovskite lattice after  $\text{Sn}^{2+}$  incorporation is believed to act as a barrier for the halide migration process. This observation was further upheld by a few other research groups also.<sup>[91], [92]</sup> However, incorporation of  $\text{Sn}^{2+}$  does employ structural instability because of ease in oxidation to  $\text{Sn}^{4+}$ . Further, Hong et al. showed the effectiveness of  $\text{Mn}^{2+}$  ions in preventing the phase-segregation of  $\text{CsPbI}_x\text{Br}_{3-x}$  perovskites.<sup>[92]</sup> There are limited reports on the influence of alloying B site cations on the phase segregation process, hence, there is a need for further exploration. Albeit the underlying reasons and mechanisms are in dark, it is well evident that alloying of A and B cations can promote the stability of MHPs.



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**Figure 2.** (a) The (200) XRD peak of an  $x = 0.6$  film before (black) and after (red) white-light soaking for 5 minutes at  $\sim 50 \text{ mW/cm}^2$ . XRD patterns of an  $x = 0.2$  film (dashed green) and an  $x = 0.7$  film (dashed brown) are included for comparison. (b) photoluminescence (PL) spectra of  $\text{MAPb}(\text{Br}_x\text{I}_{1-x})_3$  ( $x = 0.4$ ) thin-film over 45 s with 5-s increments under 457 nm,  $15 \text{ mW/cm}^2$  light at 300 K. Reproduced with permission.<sup>[60]</sup> Copyright 2015 The Royal Society of Chemistry. (c) Full Width at Half Maximum (FWHM) of the (100) XRD peak in  $\text{Cs}_y\text{FA}_{1-y}\text{Pb}(\text{Br}_{0.4}\text{I}_{0.6})_3$  perovskites with varying  $\text{Cs}^+$  content. Addition of 0.2 fraction of  $\text{Cs}^+$  results in an increased fraction of crystalline material and/or larger crystalline domains. (d) Change of the average PL photon energy  $\langle E_{\text{ph}} \rangle$  for emission from three different  $\text{Cs}_y\text{FA}_{1-y}\text{Pb}(\text{Br}_{0.4}\text{I}_{0.6})_3$  films with  $y=0.05$ ,  $y=0.20$  and  $y=0.60$ . These films are exposed to continuous illumination over the course of 30 min and excited at 400 nm with an intensity of  $100 \text{ mWcm}^{-2}$ . Reproduced with permission.<sup>[78]</sup> Copyright 2017 The Royal Society of Chemistry. (e) Predicted phase diagram of the  $\text{MAPb}(\text{I}_{1-x}\text{Br}_x)_3$  alloy. The purple and pink lines are the binodal and spinodal lines, respectively. The dashed horizontal line shows the miscibility gap at room temperature. A thermodynamically stable solid solution can be formed in the white region only. Reproduced with permission.<sup>[93]</sup> Copyright 2016 American Chemical Society. (f) Schematic Helmholtz free-energy curves in the light and dark predicting phase segregation and energy diagram sketches of the relative bandgaps of each phase. The red dashed lines show that the lowest attainable free energy in the light occurs when the material segregates into I-rich and Br-rich phases, while the blue dashed line shows that the lowest energy is attained when the material remains in a single phase in the dark. Reproduced with permission.<sup>[94]</sup> Copyright 2016 American Chemical Society. (g) Estimated free energy of formation calculated using Density Functional Theory as a function of  $x$ . The blue line/circles are generalized gradient approximation (GGA)- computed 0 K ground state formation energies, including spin-orbit coupling. The green line/squares are free energies of mixing at 300 K, assuming ideal mixing on the halide sublattice. The red

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line/inverted triangles represent the free energy difference after single-photon absorption. All energies are reported per two formula units. Blue and pink shaded areas indicate positive (phase separation favored) and negative (phase separation disfavored) free energy regions. Reproduced with permission.<sup>[69]</sup> Copyright 2017 Springer Nature. (h) Evolution curves of band gap for triple-halide films with increased ratio of Cl/(I+Br+Cl), based on Cs<sub>25</sub>Br<sub>15</sub> (red), Cs<sub>25</sub>Br<sub>30</sub> (green), and Cs<sub>25</sub>Br<sub>40</sub> (blue), respectively. The dashed line delineates the single-phase and double-phase zones corresponding to band gap raising and bandgap reducing, respectively. (i) Shifting and splitting of XRD (100) peaks show the transition from single-phase triple-halide alloy to double-phase segregation in Cs<sub>25</sub>Br<sub>40</sub> perovskites with increased Cl<sup>-</sup> content. (j) The shift of the spectral centroids of control films over time. The redshift becomes more obvious under higher injection. Reproduced with permission.<sup>[95]</sup> Copyright 2020 Science.

### 3.2. Anion variation:

To understand the fundamental mechanism of the uncanny phase segregation process, considerable efforts have been made. Through theoretical calculation, Brivio et al. proposed the phase diagram which depends on the amount of Br<sup>-</sup>/I<sup>-</sup> components under different temperatures.<sup>[68]</sup> They predicted that there exists a critical temperature (~343 K) above which MAPbBr<sub>x</sub>I<sub>1-x</sub> MHP is stable for any composition of Br<sup>-</sup> and I<sup>-</sup>. However, below this critical temperature, the MHP is preferred to have the segregated domains if Br content in the perovskite is between 15% and 75% that referred to as the miscibility gap as shown in figure 2e. The lattice mismatch of ~6% between the MAPbI<sub>3</sub> and MAPbBr<sub>3</sub> crystals is considered to have a significant influence on the instability of iso-valent solid solutions.<sup>[93]</sup> Friend and co-workers have demonstrated thermodynamically preferred segregation in consequence of lattice mismatch. They presented a shift in emission peak of MAPbBr<sub>1.2</sub>I<sub>1.8</sub> from 625 nm to 740 nm after two weeks even in the absence of photoirradiation to support the occurrence of phase

[Type here] segregation.<sup>[96]</sup> However, this model does not explain the reversibility of the mixing process. Slotcavage et al. also have put their view in favor of the thermodynamic origins of the light-induced phase segregation by calculating Helmholtz free-energy for a series of  $\text{Br}^-/\text{I}^-$  compositions in the light and dark as shown in figure 2f.<sup>[94]</sup> It can be seen from figure 2f that under illumination (red line) segregated domains have the lowest free energy compared to the mixed-phase. Contrarily, in dark conditions (blue line), the mixed-phase has the lowest free energy. In addition, Draguta and colleagues have subsequently suggested that bandgap differences between I-rich perovskite domains and the parent mixed halide perovskite are the thermodynamic drivers of free energy differences that favor phase segregation. Figure 2g illustrates that negative mixing free energies for  $\text{MAPbI}(\text{I}_{1-x}\text{Br}_x)_3$  (green line) become positive under illumination (red line) for the composition range  $x > 0.1$ . Hence, photo-segregation is preferred at  $x > 0.1$ .<sup>[69]</sup>

The halide composition of MHPs has a significant influence on halide segregation behavior. It is well known that perovskites with 1:1  $\text{Br}^-/\text{I}^-$  ratios are more prone to segregation, and perovskites having more iodide fractions are reported as more stable than their bromide counterpart.<sup>[88]</sup> Hence, the halide segregation process can be prevented by pushing the halide ratio into a stable compositional regime. Conversely, this process limits the range of bandgaps attainable. However, Bush et al. have shown that by carefully tuning both the ratio of  $\text{Cs}^+$  to FA in the A-sites and the  $\text{Br}^-$  to  $\text{I}^-$  ratio in the X-sites, a range of perovskites could be fabricated with a given bandgap, but with different photo-stability. Here,  $\text{Cs}_{0.25}\text{FA}_{0.75}\text{Pb}(\text{Br}_{0.2}\text{I}_{0.8})_3$  and  $\text{Cs}_{0.4}\text{FA}_{0.6}\text{Pb}(\text{Br}_{0.3}\text{I}_{0.7})_3$  with bandgap 1.68 eV and 1.75 eV, respectively were found to be stable against halide segregation when illuminated. Such careful compositional tuning highlights the importance of stoichiometric engineering of  $\text{X}^-$  ions in controlling the photostability of MHPs.<sup>[97]</sup>



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While the mixed iodide and bromide system is extensively studied, the influence of  $\text{Cl}^-$  on the segregation behavior of MHPs needs to be investigated. Few reports demonstrated suppression of the halide segregation process upon the incorporation of  $\text{Cl}^-$  in the MHPs lattice. Rybin et al. have shown that the incorporation of a small amount of  $\text{Cl}^-$  in  $\text{MAPbBr}_{3-x}\text{Cl}_x$  lattice suppresses the migration of  $\text{Br}^-$  ions.<sup>[98]</sup> It is also well known that because of higher electronegativity,  $\text{Cl}^-$  forms stronger ionic bonds with  $\text{Pb}^{2+}$  compared to  $\text{Br}^-$ . This provides sufficient coulomb energy to overcome the strain energy that arises due to the size mismatch between the halide atoms.<sup>[68], [99]</sup> Thus, at a lower concentration limit (up to 25 mol%), the solid solution of anions shows increased stability compared to the phase-separated constituents of  $\text{MAPbBr}_3$  and  $\text{MAPbCl}_3$ . The entropy of mixing which arises due to anion mixing stabilizes the mixed anion lattices and bestows prominent thermodynamic stability to  $\text{MAPbBr}_{3-x}\text{Cl}_x$  at room temperature. The activation energy for  $\text{Cl}^-$  (0.22-0.23 eV) is also found to be smaller than that for  $\text{Br}^-$  (0.47-0.59 eV), and the migration barrier for  $\text{Br}^-$  markedly increases with the inclusion of  $\text{Cl}^-$  ions from 0.33 eV for  $\text{MAPbBr}_3$  to 0.59 eV for  $\text{MAPbBr}_{2.75}\text{Cl}_{0.25}$ . This indicates limited inter-diffusion of halides in  $\text{Cl}^-$  mixed  $\text{Br}^-$  perovskites. A similar trend is reported for Cs-based perovskites.<sup>[100]</sup> Here,  $\text{CsPbBr}_{3-3x}\text{Cl}_{3x}$  ( $x = 0-1$ ) is easily formed by inter-diffusion of Cl and Br ions between  $\text{CsPbBr}_3$  and  $\text{CsPbCl}_3$ . However, the inter-diffusion of  $\text{I}^-$  and  $\text{Br}^-$  between  $\text{CsPbBr}_3$  and  $\text{CsPbI}_3$  is very slow. The miscibility gap between  $\text{CsPbI}_3$  and  $\text{CsPbBr}_3$  phases can be considered as the reason for the slow inter-diffusion of  $\text{Br}^-$  and  $\text{I}^-$ . Since  $\text{CsPbBr}_3$  and  $\text{CsPbCl}_3$  are completely miscible, inter-diffusion between  $\text{Cl}^-$  and  $\text{Br}^-$  is favored by entropy leading to the formation of  $\text{CsPbBr}_{3-3x}\text{Cl}_{3x}$  ( $x = 0-1$ ) MHPs.<sup>[99]</sup> It needs to be noted that  $\text{CsPbBr}_{3x}\text{I}_{3-3x}$  perovskites with high  $\text{I}^-$  content are structurally unstable.<sup>[101]</sup>

The triple halide system shows much complex behavior. It was shown that upon addition of a small amount of Cl ions, the bandgap of  $\text{FA}_{0.75}\text{Cs}_{0.25}\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3$  perovskites increases

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initially, but when the molar ratio of Cl/(I+Br+Cl) is above 10%, the bandgap of the films began to decrease (Figure 2h). This increasing and then decreasing trend of the bandgap with triple-halide alloying is anomalous and distinctly different from the monotonic growth trend reported previously in I/Br MPHs. Structural evolution observed with XRD (Figure 2i) revealed that the bandgap decreases at high  $\text{Cl}^-$  content was brought about by phase segregation into two perovskite phases. At the transition point, where the bandgap of perovskites begins to diminish with the increase in  $\text{Cl}^-$  content, the diffraction peaks of a segregated phase appeared. With the growth of the segregated perovskite phase, the lattice of the host perovskites expanded as indicated by the shift of XRD peaks to a lower angle. This is consistent with the observed band gap decrease and suggests the segregation of a high-Br/Cl phase while leaving behind I-rich lower-bandgap host perovskites. However, it was observed that an increase in Br content provides high tolerance against phase segregation (Figure 2j and 3a). This finding verified that  $\text{Br}^-$  reduced the alloy barrier between I and Cl phases for triple-halide alloys.<sup>[95]</sup> Kamat et al. have also demonstrated the effectiveness of chloride ions in preventing photo-induced halide ion segregation in MHP (Br/I) films.<sup>[51]</sup> The fraction of halide segregation decreases from 45% to 20% with an increase in the  $\text{Cl}^-$  concentration from 0% to 5% in the MHPs (Br/I). They have observed a five-fold decrease in the segregation rate constant which indicates decreased mobility of halide ions upon  $\text{Cl}^-$  incorporation. Moreover, upon the incorporation of  $\text{Cl}^-$ , the activation energy barrier for halide ion migration under illumination increases by  $\sim 4$  kJ/mol compared to Br/I MHPs. Therefore, the incorporation of a small amount of  $\text{Cl}^-$  in MHPs (Br/I) confers a significant improvement in their stability and performance in solar cells.<sup>[52, 102-105]</sup> This may be the result of several beneficial influences of chloride ions in controlling the crystallinity and morphology of perovskites.<sup>[106]</sup> Moreover,  $\text{Cl}^-$  ions have the capacity to enhance charge carrier mobility and lifetime by altering the perovskites lattice parameter.<sup>[107]</sup>

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Hence, an in-depth understanding of the  $\text{Cl}^-$  ion's role in suppressing phase segregation will assist to design perovskite solar cells with improved performance and long-term stability. The dependence of the segregation rate on the halide compositions in these studies discerns few factors dictating phase segregation kinetics.

#### 4. Polaron and defect induced halide migration

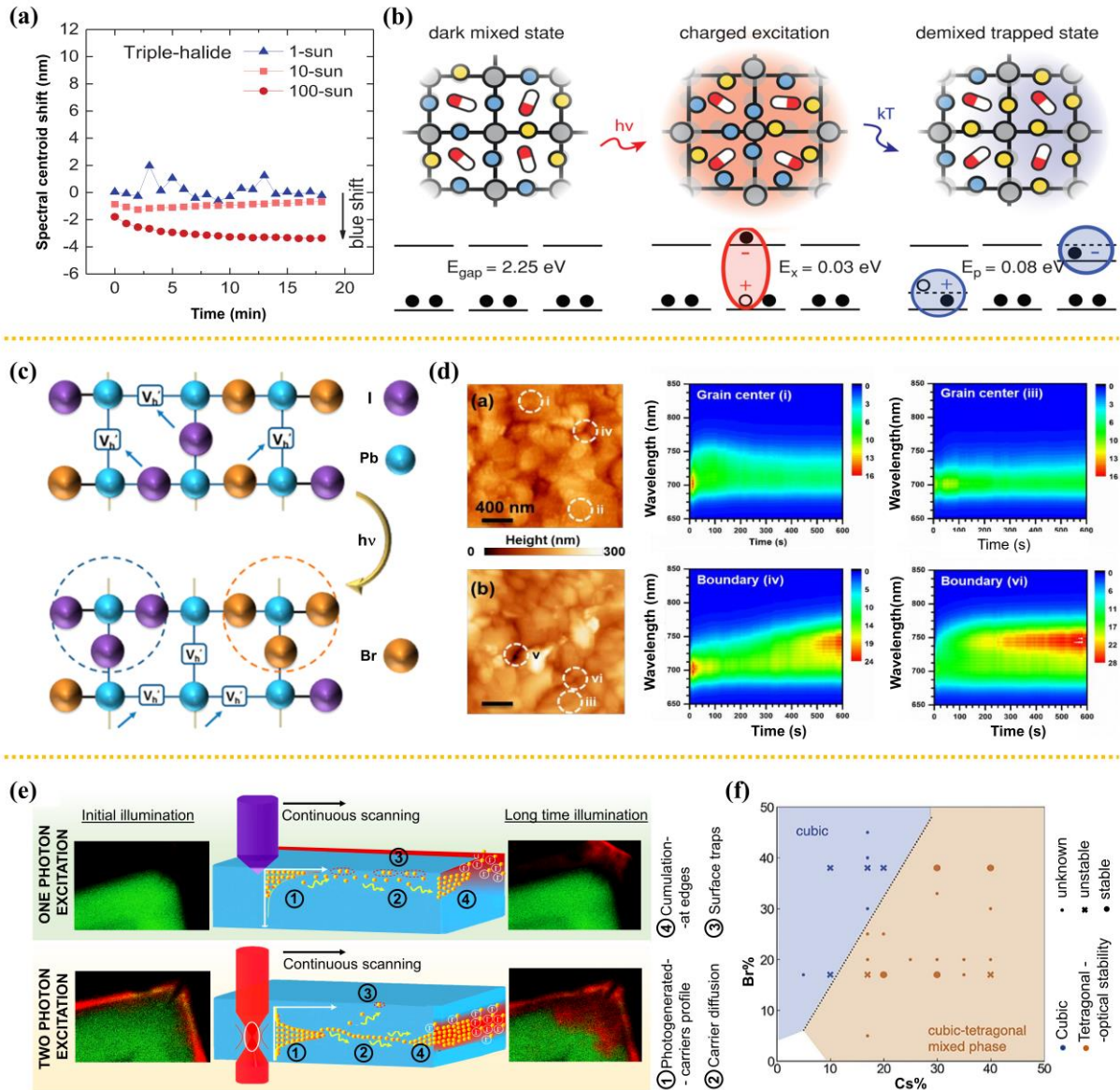
The importance of injected free carriers in the light/bias induced phase-segregation process was reported by Braly et al.<sup>[108]</sup> and Bischak et al..<sup>[109]</sup> With the help of cathodoluminescence (CL) mapping and molecular dynamics (MD) simulations, Bischak et al. provided a picture of the phase-segregation process shown in figure 3b. They have demonstrated that photoinduced phase separation occurs when charged excitations generate sufficient lattice strain in the solid solution and destabilize it. Under illumination, weakly bound e–h pair dissociates into free carriers. The ionic nature of the perovskite matrix allows the interaction of these free carriers with ions of the perovskite through the electron–phonon coupling which induces stress that deforms the surrounding lattice, and it was observed that under illumination small iodine enriched clusters localize near grain boundaries. The charge and the lattice deformation field together are termed as ‘polaron’. This polaron-induced lattice distortion influences the magnitude of the free energy as a function of halide composition. It is in match with the studies conducted by Slotcavage et al. as explained in the last section where under illumination two minima were observed and only one-minima was observed in dark indicating two-phase and single-phase existence of perovskites, respectively.<sup>[94]</sup> Their models suggest that phase segregation occurs to negate the polaron-induced lattice strain. In addition, the accumulation of polarons fosters the growth and stabilization of the segregated domains located predominantly at grain boundaries.

Previous studies have established a link between ion vacancies (defects) and halide migration. In MHPs, migration of iodide ions via ion vacancies is the main root cause for the formation of segregated domains, that attributes to the low activation energy for halide ion

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migration compared to that of A or B site cations.<sup>[80], [109]</sup> In the case of  $\text{MAPb}(\text{I}_{1-x}\text{Br}_x)_3$  perovskites, the estimated activation energy for  $\text{I}^-$  (0.58 eV) was found to have favorable activation energy for ion diffusion as compared to  $\text{Pb}^{2+}$  (2.31 eV) and MA (0.84 eV).<sup>[110]</sup> Figure 3c shows a model depicting halide ion migration in halide-deficient perovskites resulting in the formation of iodide-rich and bromide-rich domains.<sup>[111]</sup> Migration of halide ions occurs through a series of small steps between neighboring vacancies and prompts local phase segregation. Therefore, the use of excess halide ions causes reduction in halide vacancies and suppresses halide migration timescales as there is increased availability of halide ions in the lattice. However, the use of excess halide ions (Br and I) makes the material more bromine rich since bromides are bound to  $\text{Pb}^{2+}$  more strongly than iodides.<sup>[112]</sup> Hence, the use of excess halide ions cannot go beyond a certain limit in order to prevent the formation of bromide-rich compounds. Moreover, the incorporation of excess halides can alter the material properties of perovskites. Owing to the favorable formation energies, defects are always present in the perovskite materials though their number can be significantly reduced. Hence, vacancy-mediated halide ion migration cannot be eliminated completely by excess halide ions. It is worth mentioning that, number of defects directly depends on the size of MHPs crystal and the crystallinity of the MHP films. Therefore, control over the size and crystallinity of MHPs can potentially affect the phase segregation process.

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**Figure 3.** (a) The shift of the spectral centroids of triple-halide perovskites over time. The blue shift becomes more obvious under higher injection level. Reproduced with permission.<sup>[95]</sup> Copyright 2020 Science. (b) Schematic of photoinduced polaron trapping and energy scales associated with phase separation. Yellow and blue spheres represent  $\text{I}^-$  and  $\text{Br}^-$ , respectively. Reproduced with permission.<sup>[109]</sup> Copyright (2017) American Chemical Society. (c) Schematic representation of halide vacancies in the Pb-I plane of MHP films which influences halide ion migration and ultimately leads to I-rich and Br-rich domains (marked by dashed circles). Reproduced with permission.<sup>[111]</sup> Copyright (2017) American Chemical Society. (d) Local photoluminescence changes over time. (1 and 2) Topography of  $\text{CH}_3\text{NH}_3\text{PbBr}_{0.9}\text{I}_{2.1}$  perovskite.

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PL evolution over time at different regions marked in 1 and 2 at 636 nm excitation wavelength with the intensity of 500 W/cm<sup>2</sup>. Reproduced with permission.<sup>[113]</sup> Copyright (2018) American Chemical Society. (e) Schematic of the carrier transports and photoinduced phase segregation process under one-photon and two-photon illumination. Reproduced with permission.<sup>[114]</sup> Copyright (2019) John Wiley and Sons. (f) Empirical phase diagram as a function of Br- and Cs-Substitution and corresponding optical stability. Samples, where any tetragonal phase is present, are marked “tetragonal”. A rough phase boundary (dotted line) between the cubic and mixed-phase regions is included. Reproduced with permission.<sup>[115]</sup> Copyright (2019) Elsevier.

## 5. Dimensionality, crystallinity, and crystal structure control

As explained in the previous section, Bischak et al. have proposed that light-induced phase segregation causes iodide-enriched domains to form at grain boundaries to offset the strain created by polarons.<sup>[109]</sup> Similarly, Tang et al. have also observed the formation of iodine-enriched domains at grain boundaries.<sup>[113]</sup> They have shown that phase segregation at grain boundaries became prominent with illumination time while no segregation was observed within the center of the grains (Figure 3d). Along with the above-motivated studies, many more studies were performed on the polycrystalline perovskite films having high density of grain boundaries and established a significant influence of grain boundaries on the segregation process.<sup>[81]</sup> Though the exact reason behind the significant impact of grain boundaries on the phase-segregation process is needed to be determined, it was predicted that faster ionic movement along grain boundaries<sup>[116-119]</sup> and high trap densities at grain boundaries<sup>[120], [121]</sup> facilitate the phase-segregation process. Thus, reducing defect density (trap states) and the number of grain boundaries are expected to be detrimental to the segregation process. Defect density and the number of grain boundaries in MHPs can be controlled by increasing the grain size of perovskite crystals and their crystallinity. Therefore, larger size crystals are expected to be more photostable compared to smaller ones. As anticipated, Hu et al. reported that the photo-

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segregation process is hindered to a greater extent in larger size  $\text{MAPbBr}_{3-x}\text{I}_x$  perovskites grown on PTAA in comparison with small size grain of same material grown on PEDOT:PSS.<sup>[122]</sup> Hence, it promoted a projection that decreasing grain boundaries (using bigger single-crystal perovskites) foster attenuation of the phase-segregation process.<sup>[123], [124]</sup>

Surprisingly, contrary to the expectation, phase segregation was observed in MHP microplatelets.<sup>[60], [125-127]</sup> More interestingly, while few reports demonstrated that phase-segregation occurs all over the microplatelets under illumination, but on the other hand, few others demonstrated the formation of I-enriched segregated domains exclusively on the surfaces of microplates. It is well known that illumination results in charge carrier generation and these charge carriers diffuse a long distance before being trapped by surface defects or recombine radiatively. Chen et al. have tried to disintegrate the effect of surface and bulk by exciting  $\text{MAPbBr}_x\text{I}_{3-x}$  single-crystal microplatelets with one and two photons. They have demonstrated that under one-photon excitation where penetration depth is within 100 nm from the surface, PL originates from the vicinity of the perovskite surface and not from the bulk part. Herein, with one-photon laser excitation, large numbers of charge carriers are trapped by the non-radiative defects at the top surface prior to showing up at the edges, whereas the bulk of perovskite crystals has lesser charge carrier density compared to the edges. Hence, PL from the I-enriched phase can be observed only at the edges. However, for two-photon laser illumination, photogenerated charge carriers are distributed predominantly in the interior of the crystals. Hence, the chances of surface trapping are comparatively less. The charge carriers easily diffuse and accumulate at the crystal edges. Therefore, under two-photon excitation I-phase started appearing at the crystal edge and gradually migrate towards the interior of the microplatelet as PL from the I-phase was initially observed at the edges and later spread into the interior region (Figure 3e). The intensity of PL from the Br phase also decreases noticeably at the same positions indicating substitution of Br by I.<sup>[114]</sup> The corners of the platelets are the most favorable area for phase segregation, which indicates that edge states are promoting phase



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segregation in mixed halide single-crystal perovskites.<sup>[105]</sup> Because of the ability to diffuse over a long distance, a higher density of charge carriers is expected to present at the edge of crystals. The crystal edges serve as a dam of diffusing carriers. The high density of charge carriers at the crystal edges results in a higher density of phonons and lattices with high potential energy.<sup>[109], [128, 129]</sup> This leads to a high probability of the lattice halides activation and formation of interstitials and vacancies. It is already proven that these activated interstitials and vacancies mediate the phase segregation by halide substitution.<sup>[111]</sup> Based on these observations we can infer that reducing grain boundaries by increasing crystal size does not effectively diminish the phase-segregation process. However, as a great body of work evident the importance of grain boundaries in the phase-segregation process, more studies are required to unravel the significance of grain boundaries in the phase-segregation process and how phase-segregation still occurs in perovskite microplates.

Along with the crystallinity and grain size of perovskites, the crystal structure of the mixed-halide perovskite has also seemed to play an important role in the halide segregation process. MAPb(Br<sub>x</sub>I<sub>1-x</sub>)<sub>3</sub> films are found to be photo-stable at x=0.2 having the highest fraction of the perovskite in tetragonal structure. At x > 0.2, the tetragonal structure gets convert to cubic due to significant halide segregation.<sup>[60], [130]</sup> The nearest-neighbor hops available for halide ions in cubic crystals are symmetrically identical, while in a tetragonal structure there are two distinct hop pathways with different energy barriers. It was also observed that the rate of halide segregation is sensitive to the difference in the energy barriers of the two distinct migration pathways in the tetragonal perovskite structure. Thus, segregation could occur at a different rate in a tetragonal crystal lattice compared to a cubic one.<sup>[131]</sup> Meloni et al. have shown that activation energy value of equatorial-to-axial Br<sup>-</sup> vacancy migration is nearly double in β-phase MAPb(I<sub>1-x</sub>Br<sub>x</sub>)<sub>3</sub> compared to α-phase providing a kinetic rationalization for terminal x ~ 0.2 values.<sup>[132]</sup> A significant study researching the link between crystal structure and halide segregation was recently conducted by tuning of Cs<sup>+</sup> and Br<sup>-</sup> content in Cs<sub>y</sub>FA<sub>(1-y)</sub>Pb(Br<sub>x</sub>I<sub>1-x</sub>)<sub>3</sub>



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perovskite resulted in the transition from a cubic structure at  $x \sim 50\%$  and low  $y$  to a cubic-tetragonal mixed phase at low  $x$  and  $y \sim 50\%$ . While perovskites possessing cubic structure exhibited halide segregation, cubic-tetragonal perovskites showed both stable and unstable behavior (Figure 3f).<sup>[115]</sup> This suggests that crystal structure is also one of the prominent factors that govern the phase stability of MHPs. Further phase-segregation could also be seen to occur in  $\text{FAPb}(\text{Br}_x\text{I}_{1-x})_3$  MHPs.<sup>[60, 79]</sup> Unlike  $\text{MAPb}(\text{Br}_x\text{I}_{1-x})_3$ ,  $\text{FAPb}(\text{Br}_x\text{I}_{1-x})_3$  MHPs are unable to form crystalline phases in the composition range of  $0.3 < x < 0.5$ .<sup>[79, 133]</sup> This observation provides an inference that crystal structure alone cannot be considered as a major influencer of the phase-segregation process.

In addition, Draguta et. al. have proposed strategic engineering of carrier diffusion lengths and injected carrier densities that can minimize the phase segregation in MHPs. They demonstrated that reduction in diffusion length of the photogenerated carriers kinetically prevents the phase separation in  $\text{MAPb}(\text{I}_{1-x}\text{Br}_x)_3$  films. They further tested the prediction using  $\text{CsPb}(\text{I}_{0.5}\text{Br}_{0.5})_3$  thin films and their complementary nanocrystal-based films. In nanocrystals, charge carrier diffusion lengths are shorter as grain boundaries suppress charge transportation. In photoluminescence spectroscopy studies, under illumination  $\text{CsPb}(\text{I}_{0.5}\text{Br}_{0.5})_3$  thin films showed redshifts in spectra similar to their  $\text{MAPb}(\text{I}_{1-x}\text{Br}_x)_3$  counterparts due to phase separation. Contrarily, under the same excitation conditions, no redshift was observed in the case of  $\text{CsPb}(\text{I}_{0.5}\text{Br}_{0.5})_3$  nanocrystal-based films.<sup>[69]</sup> Moreover, since principle drivers of photoinduced halide segregation are charge accumulation and carrier trapping at perovskite surfaces, in perovskite nanocrystals effective surface passivation by long-chain organic ammonium capping layers used to confine the size of the nanocrystals is also one of the attributes to enhanced stability against photoinduced halide segregation.<sup>[134]</sup> Similarly, in the case of 2D perovskites, where the inorganic frameworks are separated by the organic spacers, Cho et al. have investigated halide ion mobility by tracking the changes in the bromide and iodide ions composition in physically paired films having different perovskite inorganic framework layer numbers.<sup>[124]</sup> Phase-

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segregation is suppressed in low dimensional perovskites as the diffusion coefficients of halide ions are lower in perovskites having fewer inorganic layers intercalated between organic spacer ligands.<sup>[135-137]</sup>

## 6. Influence of illumination condition

As we were discussing from the beginning, the phase segregation process predominantly occurs under illumination. Solar cells made of MHPs receive different intensities of solar radiation depending on their position, design, and weather conditions. Hence, it is important to understand how the segregation process in MHPs gets affected by properties of the illumination, i.e.  $I_{\text{exc}}$ ,  $\lambda_{\text{exc}}$ , and duty cycle (corresponding pulsed laser repetition rate,  $R_{\text{exc}}$ ) in order to have more efficient photovoltaics.

Following the expectation, a few investigations have shown that an increase in illumination intensity makes the halide segregation process faster.<sup>[72, 108, 109, 111, 131, 138, 139]</sup> It is evident that charge carriers are responsible for the segregation process as the number of charge carriers increases with an increase in the intensity of light.<sup>[108, 140]</sup> Though segregation occurs at a faster rate under higher illumination intensity, the relationship doesn't follow a linear trend. Rather, most of the studies showed an exponential saturation of the segregation process with  $I_{\text{exc}}$ . Gautam et al. have investigated the influence of illumination intensity on the phase-segregation process of  $(\text{MA}_{0.17}\text{FA}_{0.83})_{0.95}\text{Cs}_{0.05}\text{Pb}(\text{I}_{0.83}\text{Br}_{0.17})_3$  perovskite film. They have shown that  $I_{\text{em}} \propto \text{Pb}^{3/2}$  for power densities up to  $380 \text{ mW cm}^{-2}$ .<sup>[141]</sup> This superlinear dependency suggested that photogenerated carrier trapping rivals the bimolecular e-h recombination process.<sup>[142, 143]</sup> At higher excitation power-density, the power-law changes from superlinear to sublinear owing to the fact that the trap-mediated recombination process is dominated by bimolecular recombination process controlled with auger losses. Therefore, at higher excitation power-density, the saturation in  $I_{\text{em}}$  observed is attributed to the saturation of radiative bimolecular recombination centers as Auger losses start playing a key role in the recombination process.<sup>[144]</sup>

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However, there exists a material-dependent minimum threshold intensity above which the segregation process commences.<sup>[69, 79, 138]</sup> For example no segregation was observed for MAPb(Br<sub>0.4</sub>I<sub>0.6</sub>)<sub>3</sub> thin film under 0.5 mW cm<sup>-2</sup> illumination for 3 hours. Draguta et al. also observed that no segregation occurred in the MAPb(Br<sub>x</sub>I<sub>1-x</sub>)<sub>3</sub> films when the illumination was kept below 0.04 mWcm<sup>-2</sup> (Figure 4a). The rate of phase segregation increases with an increase in excitation intensity as seen through the correspondingly faster exponential growth of  $I_{\text{iodide}}/I_{\text{sat}}$  (Figure 4b). The corresponding phase separation rate constants range from  $k_{\text{forward,em}} = 0.092 - 0.823 \text{ s}^{-1}$  for  $I_{\text{exc}} = 0.26 - 58.2 \text{ mW cm}^{-2}$ . Herein,  $k_{\text{forward,em}}$  saturates at  $0.707 \text{ s}^{-1}$  when  $I_{\text{exc}}$  exceeds a critical excitation intensity of  $20 \text{ mWcm}^{-2}$ . They have explained the  $I_{\text{exc}}$ -dependent phase separation kinetics by considering the segregation process to occur by photogenerated carriers diffusing to existing iodide-rich domains on length scales comparable to electron/hole diffusion lengths (ca.  $l_{\text{e/h}} = 100 \text{ nm}$ ). The scenario is depicted in figure 4c. As with high excitation intensity carrier generation occurs in presence of multiple segregated iodide-rich domains owing to the slow phase-separation recovery. Therefore, the probability for carriers to find such a domain within a geometric volume defined by  $l_{\text{e/h}}$  increases with an increase in excitation intensity.<sup>[69]</sup> Ruth et al.<sup>[131]</sup> and Elmelund et al.<sup>[138]</sup> have also shown the existence of an intensity threshold. In the former case,  $I_{\text{exc,threshold}} \sim 100 \mu\text{W cm}^{-2}$  ( $\lambda_{\text{exc}} = 405 \text{ nm}$ ) is needed to have segregation in MAPb(I<sub>0.6</sub>Br<sub>0.4</sub>) film, while the later research revealed the dependence of threshold intensity on the temperature where threshold intensity increases with temperature. They have shown an increase in excitation intensity thresholds of MAPbBr<sub>1.5</sub>I<sub>1.5</sub> MHPs from  $30 \mu\text{W/cm}^2$  to  $100 \mu\text{W/cm}^2$  when the temperature was increased from 23 °C to 90 °C. This temperature dependency can be explained by considering the offsetting effects of halide segregation under illumination and the entropic diffusion of ions against the formed concentration gradients (explained in the following section).

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The influence of the intensity of light on the segregation process is also dependent on the halide composition of the parent perovskites. Ruan et al. have investigated a series of  $\text{MAPbI}_{3-x}\text{Br}_x$  perovskites with  $x = 0.5, 1.0, 1.5, 2.0, 2.5$  under various illumination intensities as shown in figure 4d (i-v).<sup>[145]</sup> Generally, for a specific  $\text{MAPbI}_{3-x}\text{Br}_x$  film upon increasing laser power, the PL peak at 530 nm (attributed to Br-rich regions) increases in concert with a decrease in the emission at 755 nm. This trend is more obvious for films with a low I ratio as it requires a lower laser power and shorter irradiation time to degrade the I-rich regions and observe emission from Br-rich regions.

Baker et al. have demonstrated that halide segregation dynamics are also affected by the light absorption profile and the resulting profile of the photoexcited charge-carrier density through a thin film.<sup>[80]</sup> They have investigated the segregation process in  $\text{MAPb}(\text{Br}_{0.6}\text{I}_{0.4})_3$  films and obtained the following results. Firstly, the rate of the segregation process was found to be dependent on the wavelength of the illumination. A higher wavelength of 635 nm induces much slower segregation compared to 532 nm in MHPs films. Secondly, it was also shown that the changes observed in the PL dynamics as halide segregation progressed were somewhat reversed if the perovskite film was illuminated from both sides, i.e. more uniform illumination throughout the film. This indicates the segregation dynamic may potentially be affected by the light absorption profile. They have also shown the dependence of the halide segregation process on the film thickness. The halide segregation was found to be sluggish in thin perovskite films compared to thicker ones. Thickness dependencies of the segregation process have also been verified by Hu and colleagues. They have reported photo segregation in  $\text{MAPb}(\text{I}_{0.73}\text{Br}_{0.27})_3$  film of thickness 540 nm, whereas comparatively thinner film of 320 nm remained in mixed state ( $\lambda_{\text{exc}} = 532 \text{ nm}$ ,  $I_{\text{exc}} = 100 \text{ mW cm}^{-2}$ ).<sup>[122]</sup>

Recent studies about the mechanics of the phase segregation effect have ascribed origin to polaron formation or a miscibility gap of the solid mixed halide solution because of changes in the Helmholtz free energy upon illumination.<sup>[68], [94], [109]</sup> However, both of the mentioned

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factors failed to explain the dependence of the phase segregation process on illumination conditions and film thickness. This sheds light on the role of an internal gradient in light intensity (carrier generation rate) causing halide segregation. Earlier studies have demonstrated the importance of the ion migration process in internal gradient.<sup>[146]</sup> Under illumination, halide ions migrate and move away from illuminated regions in perovskite films, and ion drift velocities in thin films have been shown to increase substantially. Besides, it needs to be noted that carriers are generated at the illuminated surface.<sup>[80]</sup> Therefore, halide ions near the illuminated region prefer to move down the film via halide defect sites. Here, in MAPb(Br<sub>1-x</sub>I<sub>x</sub>)<sub>3</sub>, as I<sup>-</sup> migrates slowly compared to Br<sup>-</sup>, I-enriched regions form near the illuminated area as Br<sup>-</sup> is depleted from the surface. These I-enriched regions due to their preferable band position act as a recombination center for photo-generated carriers generating the observed sub-bandgap PL.<sup>[147]</sup> As it was shown that the internal light profile within the perovskite has significantly influenced the phase stability of the MHPs. Therefore, it is possible to generate photostable perovskite solar cells by utilizing a back-reflecting cell design with careful control on the device layer thicknesses and absorption properties to mitigate halide segregation.<sup>[148]</sup>

Baker et al. have demonstrated that the excitation duty cycle is also one of the factors influencing the phase segregation process.<sup>[80]</sup> They have observed photo-segregation in MAPb(Br<sub>1-x</sub>I<sub>x</sub>)<sub>3</sub> films ( $x = 0.4$  and  $x = 0.6$ ) only with high duty cycle excitation. This is corroborated by Yang et al. report.<sup>[149]</sup> They have investigated the photo-segregation process in the MAPbBr<sub>1.7</sub>I<sub>1.3</sub> films with the help of PL and photomodulation (PM) spectroscopy. Under continuous wave (CW) or 10 ns pulsed laser with high repetition rates, the initial PL  $\sim 640$  nm ( $\sim 1.94$  eV) peak position shifted to a longer wavelength of  $\sim 750$  nm ( $\sim 1.65$  eV) after illumination. However, with the same 10 ns pulsed laser with lower repetition rates ( $< 500$  Hz), no shift in the PL peak position was observed irrespective of the intensity of excitation light is used. Finally, a two-step mechanism for the photo-segregation process (Figure 4e) was

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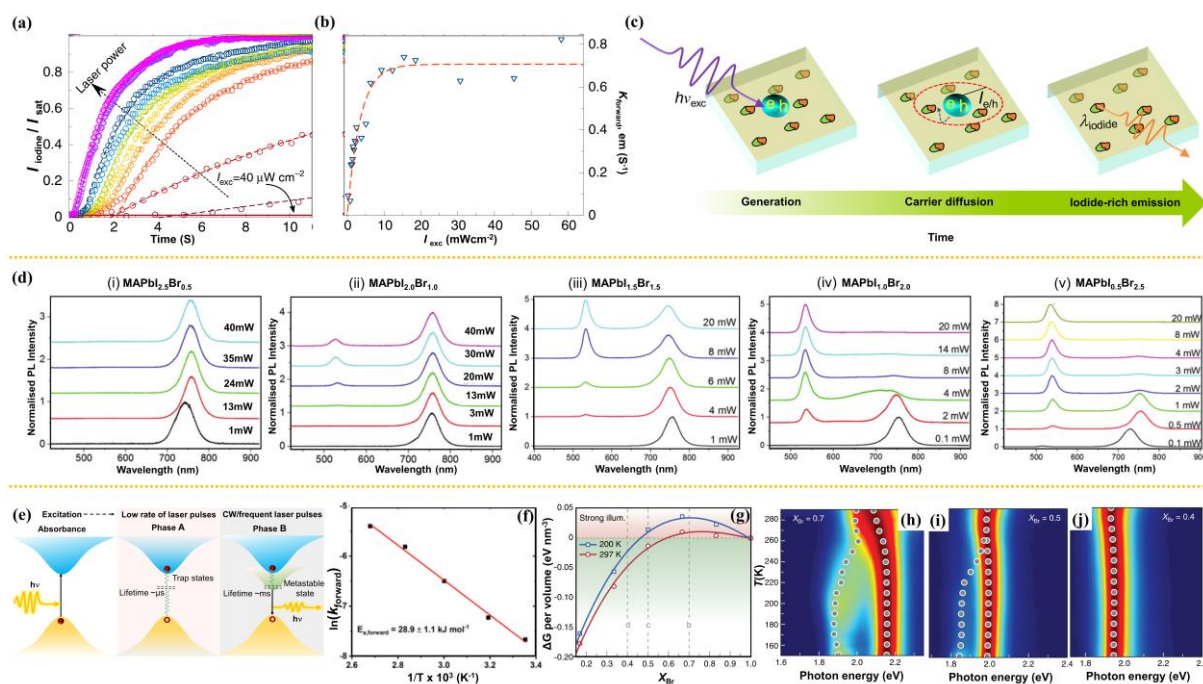
proposed: initial photon absorption will generate trap states within the bandgap. However, these traps do not lead to PL emission because of having the lifetime in the order of 1  $\mu$ s (Phase A). If subsequent incident photons are absorbed by these traps within the lifetime, it transforms into a long-lived ( $\sim$ ms) metastable state (Phase B), which behaves as a new energy band and can lead to PL emission. If those traps would have existed in films naturally, no difference in PL was expected, whether the excitation is caused by pulsed laser or by CW. Hence, these traps created by illumination (later recognized as polaronic states) are to drive ion migration.<sup>[149]</sup>

However, Knight et al. have shown that the apparent decrease in halide segregation with decreased pulse repetition rate in the experiments conducted by Yang et al. is solely related to the concomitant increase in initially injected charge carriers.<sup>[139]</sup> A decreased repetition rate requires a higher pulse fluence (higher instantaneous intensity) to maintain the same fixed average intensity value. This in turn increase the charge carrier density injected with each pulse. At low carrier densities, monomolecular trap-mediated charge recombination dominates, while radiative bimolecular band-to-band recombination becomes increasingly dominant at higher densities.<sup>[144, 150]</sup> Hence at a lower repetition rate, a larger fraction of charges will recombine through radiative band-to-band recombination allowing fewer to get trapped. As the electric field arising from the trapped charge carriers is claimed to initiate the phase segregation process, decrease in the phase segregation process was observed with decreased repetition rate. The apparent frequency dependence is therefore can be claimed as an indirect effect resulting from changes in charge-trapping with injected charge carrier density. They have also proved the effect of charge carrier density on the photo-segregation process by varying the exposure time. For experiments with longer exposure time, the illumination intensity was kept lower in order to photo-excite a smaller charge carrier density within the perovskite film. At low carrier densities, monomolecular trap-mediated charge recombination dominates. The efficiency of the charge carrier to initiate halide segregation (per unit carrier) appears higher at lower charge carrier density. Increased exposure time increases the proportion of charge carriers that

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recombine through trap states and hence causes enhancement in the amount of halide segregation per incident photon per unit volume. From, the above-mentioned studies, it is apparent that, the number of charge carriers generated and the distribution of charge carriers in the film influence phase-stability of perovskites. Hence, it is feasible to control the phase-segregation process by playing with film thickness and uniformity, which influence the carrier generation and distribution.



**Figure 4.** (a)  $I_{\text{iodide}}/I_{\text{sat}}$  ( $I_{\text{iodide}}$  – iodide-rich emission intensity,  $I_{\text{sat}}$  – saturation emission intensity) under different  $I_{\text{exc}}$  ( $I_{\text{exc}}$  – excitation intensity). The bottom solid red line shows  $I_{\text{iodide}}/I_{\text{sat}}$  when  $I_{\text{exc}} = 40 \mu\text{W cm}^{-2}$ . Excitation intensities for selected curves (from red to purple) are: 0.27, 0.79, 1.05, 1.46, 1.56, 2.27, 3.81, 5.04, 18.56, 57.28  $\text{mW cm}^{-2}$ . (b)  $I_{\text{exc}}$ -dependent emission-based first-order rate constant for phase separation. (c) Schematic of relevant kinetic processes during halide phase segregation. Reproduced with permission.<sup>[69]</sup> Copyright (2017) Springer Nature. (d) i-v: Normalized PL spectra of  $\text{MAPbI}_{3-x}\text{Br}_x$  samples ( $x = 0.5, 1.0, 1.5, 2.0, 2.5$ ) as a function of laser power (407 nm) after illuminated for 900 s under ambient conditions (22 °C, 40±5% RH). Reproduced with permission.<sup>[145]</sup> Copyright (2019) The Royal Society of Chemistry. (e) The schematic of two-step mechanism for light-induced transformation. (f) Plot

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of the natural log of the rate constant associated with halide segregation versus the inverse temperature of the perovskite material with the linear trend highlighting an Arrhenius relation. Reproduced with permission.<sup>[138]</sup> Copyright 2020 American Chemical Society. (g) The calculated  $\Delta G_{\text{light}}$  indicates that, under strong illumination, the perovskite in the composites would be less photo-stable as the temperature reduces from 297 K to 200 K. The vertical dash lines correspond to the compositions of samples examined experimentally in (h)-(j). (h)-(j): Photoluminescence spectra of  $\text{CsPb}(\text{Br}_x\text{I}_{1-x})_3/\text{Cs}_4\text{Pb}(\text{Br}_x\text{I}_{1-x})_6$  composites with  $X_{\text{Br}} = 0.7$  (h),  $X_{\text{Br}} = 0.5$  (i) and  $X_{\text{Br}} = 0.4$  (j) measured at temperatures from 150 K to 290 K (increase T). Reduced photo-stability indicates by the further split of photoluminescence peaks at lower temperatures in (h) and (i). The sample with higher iodine concentration in ( $X_{\text{Br}} = 0.4$ ) (j) remained consistently stable across the entire temperature range, consistent with other measurements where I-rich perovskites appeared more photostable. Reproduced with permission.<sup>[151]</sup> Copyright (2019) Springer Nature.

## 7. Influence of temperature

Several studies have proven the substantial influence of temperature on the phase segregation dynamics and exhibited that the rate of segregation increases with an increase in temperature.<sup>[60, 80, 109, 138]</sup> To have a more profound comprehension of the photo-segregation kinetics, Arrhenius measurements have been conducted on the  $\text{MAPb}(\text{I}_{1-x}\text{Br}_x)_3$  film ( $x = 0.4$  and  $x = 0.5$ ) for a given excitation intensity.<sup>[60, 138]</sup> Obtained forward photo-segregation activation energies are in the range of  $\sim 0.27\text{--}0.28$  eV, which are consistent with previous studies of vacancy-mediated ion migration in perovskites.<sup>[131]</sup> This indicates vacancy-mediated anion transport corroborating prior suggestions made.<sup>[138]</sup> As halide segregation relies upon ion migration, it is expected that the speed of the segregation dynamics will increase with an increase in temperature. Nonetheless, entropy is considered as the main impetus for the mixing of the halide ions in MHP systems. Since the entropic contribution to the free energy of a



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thermodynamic system is enhanced by an ascent in temperature, the mixing proclivity of the halide ions does increase.<sup>[68]</sup> Hence, the photostability of the material is expected. The disparity between these two effects of temperature on halide segregation process is a thought-provoking conflict. While plenty of experimental evidences reported a faster segregation process at higher temperatures as mentioned above, the subsequent perspective, i.e., increased entropy at higher temperatures resulting in more stable perovskite materials has been observed comparatively rare. This could be partially explained by the activation energy difference between the forward segregation and backward homogenization process. Elmelund et al. have investigated the photo-segregation process in MAPbBr<sub>1.5</sub>I<sub>1.5</sub> thin film under constant photoirradiation ( $I_{\text{exc}}=20$  mw/cm<sup>2</sup>) at different temperatures i.e. 23 °C, 60 °C, and 100 °C. Using these average rate constants for the bleach and induced absorption, the Arrhenius plot of  $\ln(k_{\text{forward}})$  versus  $1/T$  (Figure 4f) shows a linear relationship.<sup>[138]</sup> The obtained activation energy for the segregation process is  $28.9 \pm 1.1$  kJ mol<sup>-1</sup>, which is lower than that determined for the homogenization of MHP films ( $E_a$ , reverse). Therefore, the lower activation barrier of photo segregation favors halide ion segregation. Henceforth, it is needed to develop viable approaches to increase  $E_{a,\text{forward}}$  at high light intensities and high temperatures.

It was also shown that the threshold intensity (at which photo-segregation occurs) increases with temperature and follows a linear relationship in qualitative agreement with theoretical studies. The underlying concept is that the additional energy from photogenerated carriers in the perovskite film influences ion migration dynamics. By balancing the probability of phase segregation and dark recovery, an expression for the excitation intensity threshold has been provided.<sup>[68]</sup>

$$I_{\text{exc, threshold}} = \left(\frac{h}{\Delta E_{\text{g, domain}}}\right) \left(\frac{h\nu}{\alpha}\right) \left(\frac{1}{\tau v_e/h}\right)$$

$\Delta E_{\text{g, domain}}$  is an energy associated with exchanging Br<sup>-</sup> with I<sup>-</sup> within phase segregated domains,  $h\nu$  is the excitation energy,  $\alpha$  is the perovskite absorption coefficient,  $\tau$  is the carrier

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lifetime and  $v_{e/h}$  is the geometrical volume related to carrier diffusion length. Assuming the carrier diffusion lengths constant, the above equation shows a linear dependency of  $I_{\text{exc,threshold}}$  on temperature. Furthermore, using appropriate values of the parameters and room temperature, the illumination intensity threshold was calculated to be  $80 \mu\text{W}/\text{cm}^2$  which is in good agreement with the experimental value. This equation also shows an inverse relationship between  $v_{e/h}$  which supports the result obtained by Draguta et al.<sup>[69]</sup>

Interestingly, Wang et al. reported a reverse relationship. They have investigated the influence of temperature on the photo-stability of  $\text{CsPb}(\text{Br}_x\text{I}_{1-x})_3$  perovskite nanocrystals embedded in a  $\text{Cs}_4\text{Pb}(\text{Br}_x\text{I}_{1-x})_6$  endotaxial matrix. The calculated  $\Delta G_{\text{light}}$  has realized that MHPs are stable at elevated temperatures under strong illumination conditions (Figure 4g). Herein, PL spectra were taken from samples with three different compositions  $X_{\text{Br}} = 0.7, 0.5, 0.4$  (Figure 4h-4j) with  $440 \text{ mWcm}^{-2}$  illumination intensity after several hours. For perovskites having higher Br content ( $X_{\text{Br}}=0.7$ ), splitting of the PL peak was observed at room temperature indicating higher iodide and bromide concentrations in the segregated domain (Figure 4h). The spectral separation between the PL peaks increases continuously with the reduction in temperature. In the case of the sample having  $X_{\text{Br}} = 0.5$  (Figure 4i), along with an existing peak, a new peak corresponds to I-rich domains emerged by decreasing the temperature from room temperature. Surprisingly, no spectral shift was observed at room temperature for this sample. The sample with the highest iodine content ( $X_{\text{Br}}=0.4$ ) was found to be photostable irrespective of the temperature (Figure 4j). Such observation can be counted as the effect of entropy change which favors a more uniform homogenous phase at elevated temperatures.<sup>[151, 152]</sup> However, the above system is comparatively complex and the role of the matrix and MHPs in the observed behavior cannot be explained independently. Later on, similar behavior was also observed for  $\text{MAPb}(\text{I}_{1-x}\text{Br}_x)_3$  MHPs at a low illumination intensity ( $1.5 \text{ mWcm}^{-2}$ ). Temperature-dependent PL studies were done on three samples with varying halide composition ( $x = 0.15, 0.24$ , and

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0.27). No phase separation was observed for  $x=0.15$  supporting the earlier reports and claiming that photo-induced phase separation was observed only when  $\text{Br}^-$  concentration  $> 0.2$ . For  $x = 0.24$  thin film, no shift in the PL spectra was observed between 77 K and 175 K, however, at 200 K a new emission peak having the same intensity appears at 746 nm along with the initial peak at 716 nm. This suggests that at 200 K charge excitations generate sufficient lattice strain to destabilize  $\text{I}^-/\text{Br}^-$  solid solution. Further, an increase in the temperature in the range of 225 K to 275 K caused an enhancement in the intensity of 746 nm peak. At 300 K and above, complete annihilation of the peak at 746 nm suggests no phase separation due to illumination.<sup>[153]</sup> Bischak et al. also showed that mixed  $\text{I}^-/\text{Br}^-$  perovskite undergoes demixing transition with a critical temperature of 190 K.<sup>[125]</sup> However, the disappearance of the photo-induced phase separation is observed at 300 K and above. Theoretical calculations also revealed an upper limit of temperature at which perovskites are stable for any composition.<sup>[68]</sup> This temperature dependency of the segregation process can be explained by understanding electron-phonon interactions. At higher temperatures, electron-hole mobility decreased because electron-phonon interactions hinder ion migration and hence phase separation. Surprisingly, the crystallographic phase transition from orthorhombic to tetragonal and from tetragonal to cubic phase for  $x = 0.24$  composition is 200 K and 280 K, respectively that are in close proximity to the formation and annihilation of photo-induced phase separation temperatures. A similar trend was followed by the sample having  $x = 0.27$  composition. Further detailed investigations are required to understand whether the crystallographic phase transition associated with the temperature change or the temperature itself drives the phase-segregation process. However, for most of the MHP having composition  $X_{\text{Br}} \geq 0.3$ , photo-induced phase segregation occurs at room temperature and remixing occurs at much higher temperatures ( $\geq 325\text{K}$ ).<sup>[72]</sup> Similarly, a low critical solution temperature of 140 K is observed in the analysis of  $\text{Br}^-/\text{Cl}^-$  perovskites.

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For  $\text{I}^-/\text{Cl}^-$  perovskites, nonetheless, the critical solution temperature (1800 K) is well above room temperature, consistent with experimental observations that such blends are unstable.<sup>[109]</sup>

As temperature change can drive the phase-segregation equilibrium in either direction, it is important to understand the conditions under which temperature drives or opposes the phase-segregation process. Since phase-segregation occurs predominantly under illumination or electric bias, it is necessary to comprehend the impact of temperature under illumination/electric bias. Generally, without illumination, an increase in temperature increases the diffusion speed of the ions and results in a thermodynamically stable mixed phase. However, under illumination, elevated temperature either promotes or opposes the phase-segregation process depending on the other conditions employed. Therefore, control over temperature can be easily achieved and directs an efficient strategy to mitigate the photo-induced phase segregation.

## 8. Atmospheric condition

The stability of perovskites under atmospheric conditions is perhaps the biggest challenge that needs to be handled.<sup>[154-160]</sup> There exist several reports displaying an increase in photoluminescence quantum yield (PLQY) of  $\text{MAPbBr}_3$ <sup>[159, 160]</sup> and  $\text{MAPbI}_3$ <sup>[154-158]</sup> perovskites films under illumination. This behavior has been attributed to a decrease in the density of electronic trap states. It was widely reported that oxygen and humidity are the major causes of perovskites degradation.<sup>[161-163]</sup> However, It has been demonstrated that oxygen molecules diffuse into the perovskite crystal and form a superoxide species ( $\text{O}_2^-$ ) when they interact with photogenerated carriers. This superoxide species fill the iodide vacancies and hence passivate potential trap states.<sup>[154-156]</sup> As discussed earlier, trap states and halide ion vacancies play a significant part in the phase segregation process. It is also well known that excited charge carriers recombine through trap states and impact the speed of halide ion movement. Accordingly, any mechanism that changes the density of trap states ought to alter the

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segregation dynamics. Furthermore, movement of the halide ions in the MHP films is thought to occur through lattice-site hopping.<sup>[110]</sup> Therefore, superoxide species that inhabit the halide vacancy sites are expected to attenuate the mobility of halide ions causing reduction in the rate of halide segregation. Fan et al. experimentally proven that light-induced phase separation can be dramatically smothered by oxygen passivation, and more excitingly the phase-segregated MHPs can be recovered to their original uniform state in an oxygen environment.<sup>[164]</sup> They have researched the phase segregation in the  $\text{MAPb}(\text{Br}_x\text{I}_{1-x})_3$  film for a series of halide compositions. Absorption and normalized PL spectra of  $\text{MAPb}(\text{Br}_x\text{I}_{1-x})_3$  films before and after illumination in pure nitrogen, moisture ( $\text{N}_2 + \text{H}_2\text{O}$ , ~35% relative humidity), and pure oxygen environments are shown in figure 5a and 5b. Under  $\text{N}_2$  and humidity, a significant shift in the PL is noticed indicating phase segregation. On the other hand, under oxygen environment, most of the compositions did not show any significant PL shift demonstrating no phase separation in those films. Nevertheless, at high bromide concentration, additional small peaks were observed indicating phase segregation for high  $\text{Br}^-$  ratio even under  $\text{O}_2$  atmosphere. Light illumination provides energy for halide migration in MHPs and simultaneously accelerates  $\text{O}_2$  infiltration into perovskite films passivating the traps existing in the film.

Density functional theory (DFT) was also employed to investigate the role of defects particularly  $\text{Pb}^{2+}$  interstitial ( $\text{Pb}_i$ ) and halide vacancies, and to understand how the oxygen atoms passivate the trap states. The passivation effects of the oxygen atom can be understood by considering the projected density of states (PDOS) as presented in figure 5c–5f. For  $\text{Pb}$  interstitial ( $\text{Pb}_i$ ) the foreign and original  $\text{Pb}$  atom form a  $\text{Pb-Pb}$  dimer sharing the center of the  $\text{PbX}_6$  ( $\text{X} = \text{I/Br}$ ) octahedron with a deep-bandgap state at ~1 eV below the CBM. The bandgap state is mainly contributed by the  $\text{Pb-6p}$  orbital of the  $\text{Pb-Pb}$  dimer (inset of Figure 5c) and serves as a deep hole trap due to the state being occupied by two excess electrons. After inserting an oxygen atom into the  $\text{Pb-Pb}$  dimer, the deep trap state in the bandgap can be fully

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passivated (Figure 5d). The oxygen atom takes two excess electrons and forms a Pb–O–Pb bond. Here, oxygen atoms are expected to impede the migration channels of halide ions. The binding energy of Pb–O is also more than Pb–Br and Pb–I owing to the higher electronegativity of O atoms. Hence, halide ions need to overcome higher diffusion barriers that help in enhancing the phase stability of MHPs. The schematic of the phase segregation process in presence of O<sub>2</sub> is shown in figure 5g. In general, under illumination, the trapped charges cause band bending and generate an electric field leading to the migration of iodide ions to the crystal surface. In presence of O<sub>2</sub>, the surface traps get passivated and the band bendings near the surface get reduced. The accumulated iodide ions are proposed to diffuse back to the bulk due to the concentration gradient resulting in uniform MHP films.<sup>[164]</sup>

Contrarily, some studies indicated enhancement in the rate of segregation upon exposure to humidity or oxygen environment. Ruan et al. have reported oxygen-induced degradation of MAPbI<sub>0.5</sub>Br<sub>2.5</sub> film.<sup>[145]</sup> These results have shown that even in a low volume O<sub>2</sub> content can trigger fast degradation of I-rich domains, and the concurrent increase in I<sup>−</sup> vacancies under high power laser irradiation. Knight et al. have also demonstrated the profound impact of atmospheric conditions on the light-induced phase segregation process.<sup>[139]</sup> Considering the PL emission peaks, they have investigated the stability of MAPb(Br<sub>0.5</sub>I<sub>0.5</sub>)<sub>3</sub> films that were maintained under, vacuum, ambient air, and pressurized (2 bar) nitrogen conditions, and also the films coated with Poly(methyl methacrylate) under vacuum. To induce repeated segregation and remixing, the films were exposed to eight illumination cycles with a single cycle consisting of 15 s of illumination followed by 30 min of darkness. Under vacuum, as the number of electronic trap states increases, an average reduction in the PL spectra was observed.<sup>[154, 160]</sup> In the case of ambient air, passivation of trap states occurs by the superoxide species causing an enhancement of the average PL signal across cycles. Finally, in films under N<sub>2</sub> conditions and with Poly(methyl methacrylate) protection layer in vacuum, neither effect was noticed. They

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have also demonstrated that under vacuum or air irreversible photoinduced changes occurred.

Humidity also plays a significant role in the photo-segregation process. Under illumination, enhancement in the segregation rate of  $(\text{Cs}_{0.05}\text{MA}_{0.16}\text{FA}_{0.79})\text{Pb}(\text{Br}_{0.17}\text{I}_{0.83})_3$  film was observed.<sup>[165]</sup> Another study made by Howard et al. registered a redshift of the PL emission peak from  $\text{Cs}_y\text{FA}_{(1-y)}\text{Pb}(\text{Br}_{0.38}\text{I}_{0.62})_3$  films when they were held under humid air environment.<sup>[166]</sup> However, the PL measurements were made with minimal light-soaking ( $0.015\text{ s}$  at  $6000\text{ mWcm}^{-2}$ ) for an exceptionally brief time frame scale. Hence, the observed change in PL (phase segregation) can be considered as a function of humidity alone. Nevertheless, PL measurement alone cannot be confirming the occurrence of phase segregation. Precise examinations are needed to unwind the impact of humidity on MHPs.

In contrast to the above-mentioned impacts of oxygen and environmental conditions on the phase-segregation process, few reports have shown no significant changes under atmospheric conditions.<sup>[113], [167]</sup> This discrepancy may be originated from the structural differences of MHPs or differences in measurement conditions. However, encapsulation of the perovskites with some protecting layer or use of appropriate ETL layer is shown to stifle the influence of surroundings effect.

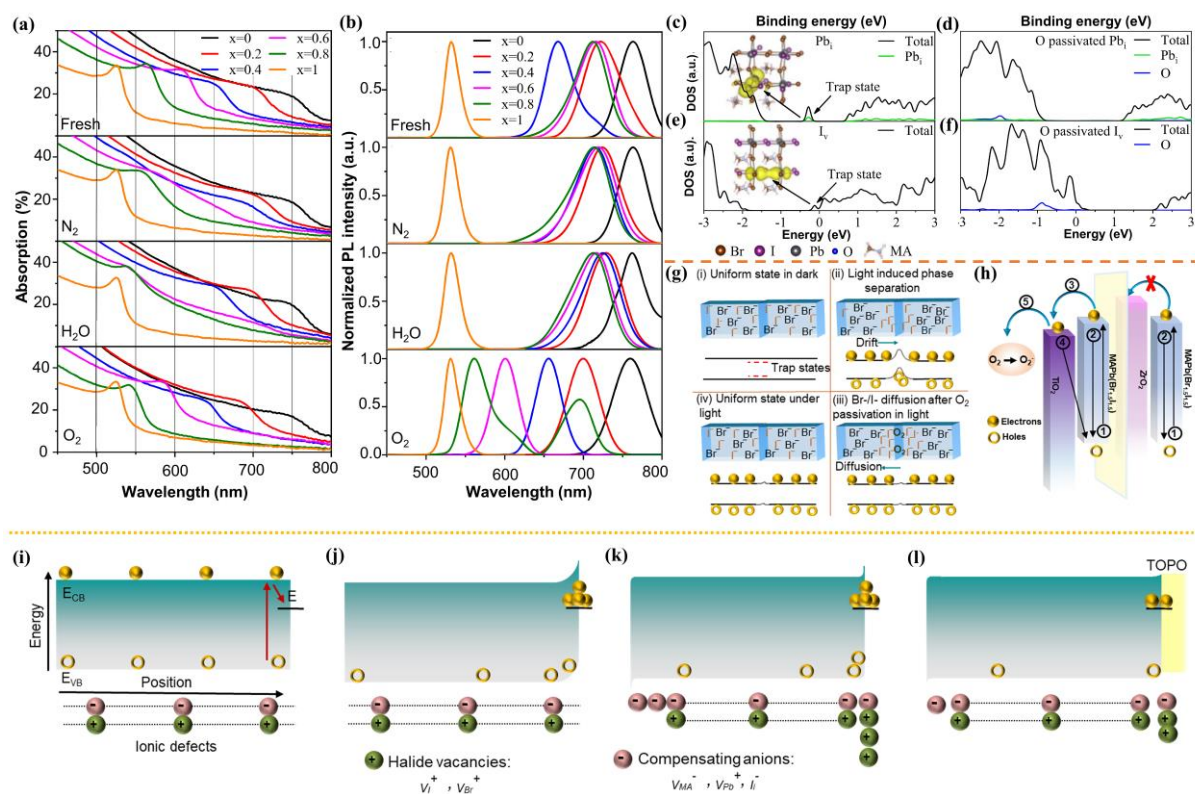
## 9. Hole assisted phase segregation and passivation

DuBose et al. have shown that light-induced halide segregation was observed when  $\text{TiO}_2$  substrates were employed as the electron transport layer (ETL) in MHPs based solar cells, while no such effect was observed if  $\text{ZrO}_2$  was employed as ETL layer.<sup>[168]</sup> So, it can be expected that  $\text{TiO}_2$  with its favorable conduction band energy facilitates the capture of electrons from excited MHPs, and thus promotes charge separation while leaving holes trapped within the MHP film (Figure 5h). Then the electrons get scavenged by the atmospheric oxygen, whereas holes cascade to iodide-rich sites of MHP as a result of band energy alignment within the segregated film. It has also been shown that selective trapping of holes at iodide sites induces lattice

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instability and moves iodide ions towards the grain boundaries through defect sites.<sup>[113]</sup> Under continuous irradiation, iodide accumulates at the grain boundaries leading to halide ion segregation. Therefore, under anodic potential and in contact with a solvent such as dichloromethane (DCM), iodide ions get ousted into the solvent leaving behind MAPbBr<sub>3</sub> film. It is worth mentioning that hole-induced phase segregation was also observed when holes are injected electrochemically into MHPs.



**Figure 5.** (a) Absorption and (b) normalized PL spectra of MAPb(Br<sub>x</sub>I<sub>1-x</sub>)<sub>3</sub> films before illumination and after illumination in pure nitrogen, moisture (N<sub>2</sub> + H<sub>2</sub>O, ~35% relative humidity), and pure oxygen environment. The PL measurement was done with continuous excitation with a wavelength of 350 nm and an intensity of 6 mW cm<sup>-2</sup>. The absorption onset of as-prepared MHPs shifts gradually from 760 to 530 nm with increasing Br content. In the presence of N<sub>2</sub> and moisture redshift of the PL peaks were observed indicating phase separation. In the O<sub>2</sub> environment, the PL peak positions, which were pinned in the infrared range, be blue-shifted to their absorption edges. This indicates that the phase separation is recovered to the



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original uniform state in those films. The PDOS of (c,d)  $\text{Pb}_i$  and (e,f)  $\text{I}_v$  before and after  $\text{O}_2$  passivation. The insets show the orbital distribution of the trap states. (g) Schematics of the reversion of phase separation in MHPs due to  $\text{O}_2$  passivation. (i) Uniform state in the dark, (ii) light-induced phase separation caused by the trapped charges at the surface, (iii) iodide diffusion back to the bulk due to the concentration gradient, and (iv) uniform state under the light. Reproduced with permission.<sup>[164]</sup> Copyright (2019) American Chemical Society. (h) Band diagram depicting the possible excited-state pathways of  $\text{MAPb}(\text{Br}_{1.5}\text{I}_{1.5})$  interacting with  $\text{TiO}_2$ ,  $\text{ZrO}_2$ , and  $\text{O}_2$ .<sup>[168]</sup> (i-l) Annotated band diagrams produced from drift-diffusion simulations highlighting the hypothesized mechanism by which surfaces encourage halide segregation for mixed-halide perovskites on glass. (i) Photo-induced occupancy of surface defect states, (j) band-bending induced by trapping, (k) halide vacancy and hole redistribution to compensate the trap-induced field, and (l) the TOPO treatment partially passivating surface states and consequently less number of charges accumulate at the perovskite surface. Reproduced with permission.<sup>[134]</sup> Copyright (2018) American Chemical Society.

Meanwhile, an ample amount of studies have established a strong connection between trap-mediated recombination of excited charge carriers with halide segregation.<sup>[140], [169, 170]</sup> By varying the intensity of illumination and controlling the total number of cumulative incident photons on MHP film, Knight et al. have correlated the fraction of charge-carriers recombined through trap states with the extent of halide segregation.<sup>[139]</sup> In addition, trap states are accountable for the majority of the preventable losses in the solar cell. Hence, the passivation of traps can be considered as a viable method to suppress the extent of phase segregation.

Among the myriad surface passivation techniques explored, the Lewis base trioctylphosphine oxide (TOPO) proffers a significant impact on light-induced phase segregation in the  $\text{CH}_3\text{NH}_3\text{PbI}_2\text{Br}$ .<sup>[134, 171, 172]</sup> Ligands possessing Lewis base character are capable of electron donation to the surface of the MHPs and hence passivate some surface defect on the perovskite. After electron donation, TOPO acquires a net positive charge and an electric

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dipole is created. Belisle et al. proposed a general model to elucidate the influence of surface treatments on the halide segregation process.<sup>[134]</sup> They have suggested that carrier-trapping at the surface of the MHPs film induces electric fields that are opposed by the dipole of the electron-donating layer. The proposed mechanism is presented in figure 5i-l. In general, before the illumination of light, it is expected that mobile halide vacancies are uniformly distributed and charge-compensated (Figure 5i). Trapping of electrons (Figure 5j) induces the formation of an electric field pointing towards the surface of the film. To shield the trap-induced field, halide vacancies migrate towards surface states and accumulate leaving a depletion region populated with immobile anions (Figure 5k). If the surface of MHP is treated with an electron donor (TOPO), the molecular dipole opposes this trap-induced field. This, in turn, prevents the accumulation of holes and vacancies in the surface of the perovskite layer (Figure 5l).

Monovalent cations are also shown to passivate MHPs effectively. Abdi-Jalebi et al. have demonstrated the prevention of phase-segregation in (CsMAFA)Pb(Br<sub>0.15</sub>I<sub>0.85</sub>)<sub>3</sub> perovskite films by the addition of potassium iodide.<sup>[20]</sup> A comparative pattern was noticed for a series of MHPs compositions. In contrary to the earlier reports where monovalent cations are shown to be incorporated in the perovskite lattice, this report evident accumulation of K<sup>+</sup> ions at the surface. Herein, excess iodide ions provided by KI passivates the trap states resulting from halide vacancies and reduce the halide ion mobility. In augmentation, potassium ions accumulate at the grain surfaces of the perovskite while passivating the trap states that are leftover and limit halide ion movement in those regions. Yang et al. also have observed suppression of the phase-segregation process in CsPbI<sub>3-x</sub>Br<sub>x</sub> perovskites by passivating with KBr.<sup>[171]</sup> As K<sup>+</sup> and Br<sup>-</sup> ions possess higher binding energy compared to Pb<sup>2+</sup> and Br<sup>-</sup> ions, migration of Br<sup>-</sup> ions is hindered and the formation of halide defects at grain boundaries is reduced. Therefore, by passivating defects at the grain boundary of the MHPs film, K<sup>+</sup> and Br<sup>-</sup> ions suppress halide ions movements and hence the phase-segregation process. Compared to

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$\text{Br}^-$  and  $\text{I}^-$ ,  $\text{Cl}^-$  vacancies create relatively deeper trap states in MHPs.<sup>[173]</sup> Zhang et al. have developed a strategy to passivate  $\text{Cl}^-$  vacancies in MHP quantum dots using organic pseudohalide (n-dodecylammonium thiocyanate (DAT)).<sup>[174]</sup>

It is known that the formation of bromide vacancies requires lower activation energy compared to iodide vacancies.<sup>[115], [175]</sup> Hence, in MHPs bromide will prefer to move away from the surface prompting the development of iodide-rich phase on the surface. Albeit only MA-based perovskites were extensively investigated, few studies suggest that in Cs perovskites the energy of the electronic traps at the surface would be modified by the shift in the valence band edge such that they are no longer deep, which would likewise reduce the driving force for halide segregation.<sup>[97, 115]</sup> Passivation of trap states on MHP surfaces can be considered as a potential method to control halide segregation. However, the passivation of MHPs without alternation of their intrinsic properties is a daunting task. Moreover, finding suitable passivating agents and incorporating them into the perovskites via an appropriate procedure is challenging.

## 10. Summary and Perspectives

MHPs are significant in tandem solar cells and light-emitting diodes (LEDs) due to their compositional diversity for tuning their electronic band structure and improvement in stability to some extent. However, the inherent ion migration character under illumination or charge-carrier injection is circumscribing the utilization of band tunability in MHPs to the fullest possible. In MHPs, halide segregation creates domains having different band gaps which lead to the funneling of excitons from higher bandgap domains to lower bandgap domains. Therefore, this leads to lowering the maximum possible open-circuit voltages and hysteresis in solar cells, and color impurity in LEDs. Moreover, this spectral instability shows an adverse effect on the performance of other optoelectronic device applications also. This perspective assessed the genesis of phase segregation caused by halide migration and discussed a few

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strategies that preclude phase segregation. In MHPs the ease with which phase segregation proceeds are either associated with structural stability or by external factors. This perspective emphasized crucial factors such as composition, crystallinity, and dimensionality of the MHP material on the segregation process. It was observed that the rate of halide segregation is sensitive to the difference in the energy barriers of the two distinct migration pathways in the tetragonal perovskite structure compared to the cubic one. Furthermore, we discussed the stability of nanocrystals and 2-D perovskites against phase segregation as a result of their reduced charge carriers' diffusion lengths.

Then we discussed the influence on ion migration in response to external stimuli such as illumination intensity, temperature, and atmospheric conditions. Here, the challenge is to understand the mechanism where in the absence of illumination, temperature drives the halide mixing to form single-phase perovskites while under illumination above the threshold intensity, there exists conflict regarding the influence of temperature on the segregation process. Though it is well evident that increase in temperature increase the phase segregation process, it is under question that how the contribution of temperature towards entropic stabilization is not a common realization? Whether the thermodynamic model needs improvement or the influence of temperature on anion mixing is not significant? Though we are keenly interested in understanding forces that drive phase segregation, dynamics of the reverse process are yet to be discovered and speeding of the recovery time may be an alternate route to solve the segregation related issues. Finally, we have mentioned in detail the impact of the hole-assisted halide migration and passivation that influence the photostability of the MHPs. Although a lot of questions remain unanswered, this perspective provides an overall idea of the underlying mechanism of the phase-segregation process. If trials are made to raise the activation energy of the photo-segregation process under a thermodynamically stable compositional window, the undesired process of segregation can be halted significantly. Hence, the chances for perovskites to stay in the business at high for long can be rocketed.

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## Conflict of interest

There are no conflicts to declare.

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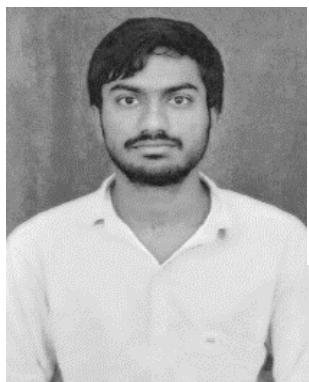
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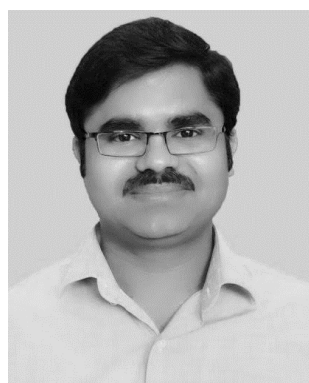
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### **The table of contents:**

Mixed-halide perovskites are tuneable photo absorbers and emitters because of their compositional dependent properties. This perspective provides a meticulous understanding of the critical views on how the mixed-halide perovskites are going to be the cutting-edge next-generation materials, focusing on the implications that have become seemingly difficult to extrapolate and predict the realistic performance of optoelectronic devices in working conditions.

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